

Clinic-radiologic predictors of pathological characteristics in pure ground-glass nodules: Development and validation of a predictive nomogram



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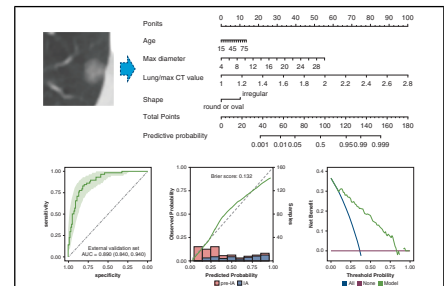
ABSTRACT

Objectives: The objective of this study was to explore the relationship between the clinic-radiologic characteristics of pure ground-glass nodules (pGGNs) and key pathologic features of lung adenocarcinoma, including tumor invasiveness, proliferative activity, driver oncogenic mutations, and the tumor immune microenvironment.

Methods: A total of 1070 surgically resected pGGNs were retrospectively analyzed and categorized pathologically into 224 precursor glandular lesions (PGLs), 600 minimally invasive adenocarcinomas (MIAs), and 246 invasive adenocarcinomas (IAs). Receiver operating characteristic curves and area under the curve were used to assess the diagnostic performance of various size and density parameters. After adjusting for clinical covariates, computed tomography (CT) features were compared across the different pathologic subgroups. The relationships between clinic-radiologic characteristics and the Ki-67 proliferation index, EGFR mutation status, and tertiary lymphoid structures (TLS) were examined. Finally, a clinic-radiologic nomogram was developed and externally validated to enable accurate preoperative prediction of IA presenting as pGGNs.

Results: Receiver operating characteristic analysis revealed that density-related parameters of pGGNs had strong discriminatory power for both MIA and IA, with the lung/max CT value showing the greatest diagnostic performance, followed by the max CT value. The optimal cut-off values for differentiating MIA from PGL were 1.62 (lung/max CT value), -548 HU (max CT value), and 8.6 mm (max diameter), whereas for distinguishing IA from MIA, the respective thresholds were 1.89, -458 HU, and 11.5 mm. After adjusting for clinical covariates, CT features such as size, density, shape, border, and voxel heterogeneity remained significantly different across the PGL, MIA, and IA groups. Furthermore, a positive correlation was found between max CT value and pathologic indicators, including Ki-67 proliferation index, EGFR mutation, and TLS. A clinic-radiologic nomogram that incorporated age, max diameter, lung/max CT value, and shape demonstrated excellent discriminatory ability for IA (area under the curve, 0.883-0.920), along with good calibration and clinical utility.

Conclusions: Size and density, as key radiologic features in the assessment of pGGNs, proved to be reliable predictors of pathologic invasiveness, proliferative activity, EGFR mutation status, and TLS levels. A clinic-radiologic nomogram for noninvasive prediction of IA offered a valuable tool to guide surveillance strategies and therapeutic decisions for patients with pGGNs. (J Thorac Cardiovasc Surg 2026;171:742-53)



Nomogram and performance of the model predicting invasive adenocarcinoma in pGGNs.

CENTRAL MESSAGE

Size and density are robust imaging biomarkers of pGGNs; a nomogram was established to stratify adenocarcinoma risk and support individualized clinical management.

PERSPECTIVE

This study highlights the diagnostic value of CT-based size and density metrics in predicting invasiveness, proliferative activity, EGFR mutations, and TLS levels in pGGNs. A validated clinic-radiologic nomogram offers a practical tool for the preoperative identification of invasive adenocarcinoma, aiding clinical decision-making and personalized management.

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Abbreviations and Acronyms

AAH	= atypical adenomatous hyperplasia
AIS	= adenocarcinoma in situ
AUC	= area under the curve
DCA	= decision curve analysis
EGFR	= epidermal growth factor receptor
EVT	= external validation set
H&E	= hematoxylin and eosin
HRCT	= high-resolution computed tomography
IA	= invasive adenocarcinoma
IQR	= interquartile range
IVS	= internal validation set
Ki-67 PI	= Ki-67 proliferation index
LDCT	= low-dose computed tomography
LUAD	= lung adenocarcinoma
mGGN	= mixed ground-glass nodule
MIA	= minimally invasive adenocarcinoma
mIF	= multiplex immunofluorescence
NSCLC	= non–small cell lung cancer
OR	= odds ratio
pGGN	= pure ground-glass nodule
PGL	= precursor glandular lesions
PSM	= propensity score matching
ROC	= receiver operating characteristic
TLS	= tertiary lymphoid structure
TS	= training set
U-HRCT	= ultra-high-resolution computed tomography
VPI	= visceral pleural invasion
WT	= wild type



Scanning this QR code will take you to the table of contents to access supplementary information.



Lung cancer remains the most frequently diagnosed malignancy and the leading cause of cancer-related death globally.¹ A recent cohort study from Taiwan that focused on patients who were never-smokers found that 77.4% (246/318) of lung cancers detected through low-dose computed tomography (LDCT) screening were stage I.² Ground-glass opacity nodules are a common radiologic manifestation of early-stage lung adenocarcinoma (LUAD) and can be classified into pure GGNs (pGGNs) and mixed GGNs (mGGNs).^{E48} pGGNs are defined as lesions that lack any solid component that would obscure the underlying lung architecture, and they account for approximately 84.9% of lung nodules identified

by LDCT-based lung cancer screening.³ Although most incidentally detected pGGNs are benign, often attributable to inflammation or focal fibroplasia, approximately 80% of persistent pGGNs are pathologically confirmed as atypical adenomatous hyperplasia (AAH), adenocarcinoma in situ (AIS), minimally invasive adenocarcinoma (MIA), or invasive adenocarcinoma (IA).^{4,E49}

Numerous studies have highlighted the slow growth pattern and favorable long-term prognosis of pGGNs; however, overtreatment remains a significant clinical concern, particularly in China.⁵ It is crucial to accurately predict and assess the pathologic characteristics and behavior of pGGNs to prevent unnecessary surgeries. Established risk factors such as advanced age and a history of smoking are known to increase the likelihood of malignancy in pulmonary nodules.^{E50} Radiographically, the size of the lesion, as determined by CT scans, plays a vital role in clinical decision-making.^{6,7} In addition, nodule density has been recognized as an important predictor of MIA and IA.^{8,9} Furthermore, other CT features, including pleural indentation, irregular shapes, spiculated or lobulated margins, bubble lucency, air bronchograms, changes in vascular structures, and voxel heterogeneity, are also associated with invasiveness.^{4,9-14} However, the practical usefulness of these malignant indicators is debated, because they might affect the clinical judgments of health care providers and could increase anxiety among patients.

The Ki-67 proliferation index (PI) is a commonly used biomarker for assessing tumor proliferation activity and has been linked to survival rates in stage I non–small cell lung cancer (NSCLC).¹⁵ The association between Ki-67 PI and the radiologic characteristics of pGGNs is not well established. In populations from East Asia, typically consisting of female, nonsmoking individuals, more than 70% of LUADs that appear as GGNs contain oncogenic driver mutations, with epidermal growth factor receptor (EGFR) mutations being the most common, occurring in more than 60% of such cases.¹⁶ Previous investigations suggest that EGFR-mutant tumors may be more prone to progress through the AAH/AIS-MIA-IA spectrum in a temporally dependent manner.^{16,17,E51} However, genetic analysis still relies mainly on surgical specimens, limiting its preoperative diagnostic value—particularly in patients with multiple pGGNs—highlighting the need for noninvasive tools for risk assessment. Tertiary lymphoid structures (TLS), ectopic lymphoid formations found within or near tumors, are gaining increasing attention as predictive biomarkers for patient survival and immunotherapeutic response in NSCLC.¹⁸⁻²¹ Although previous studies have reported a lower presence of TLS in GGNs, particularly in pGGNs, the radiologic signs associated with high TLS density have not been clearly defined.²²

In this study we conducted a thorough comparison of clinic-radiological features among precursor glandular

lesions (PGLs), MIA, and IA. It also assessed the predictive accuracy and determined the optimal thresholds for crucial CT parameters that help distinguish between levels of pathologic invasiveness. Moreover, the study examined the relationships between imaging characteristics of pGGNs and biological markers such as Ki-67 PI, EGFR mutation status, and TLSs. Finally, we developed a clinic-radiologic nomogram to accurately predict IA in pGGNs, with the goal of informing personalized surveillance and intervention strategies.

METHODS

Patients

This retrospective study received approval from the institutional review board of Renji Hospital, Shanghai Jiao Tong University School of Medicine (Shanghai, China; approval number: LY2025-005-B; approval date: January 13, 2025), and informed consent was waived. The study reviewed records of patients with pGGNs who underwent surgical resection and received a pathologic diagnosis of AAH, AIS, MIA, or IA from January 2017 to December 2021. The primary cohort consisted of cases from Renji Hospital, and additional cases from 4 external institutions, namely Qijing Traditional Chinese Medicine Hospital, The Second Hospital of Jiaxing, Qingdao No. 8 People's Hospital, and Ningbo Hangzhou Bay Hospital were used to create an external validation set (EVS).

Analysis of CT Features

High-resolution computed tomography (HRCT) scans were independently analyzed by 2 fellowship-trained thoracic radiologists at a picture archiving and communication system workstation, with both radiologists unaware of any clinicopathologic data. Each observer evaluated and measured various radiologic features, including lesion size, CT attenuation values of the lesion and surrounding lung tissue, anatomical location, number of lesions, pleural attachment, shape of the lesion, clarity of the borders, presence of air bronchograms, bubble-like lucency, vascular patterns, and voxel heterogeneity. Detailed guidelines for the measurement and interpretation of these features are available in [Appendix E1](#).

Analysis of Pathology

Surgically excised specimens were fixed in formalin and embedded in paraffin, then stained with hematoxylin and eosin (H&E). Pathologic analysis of the slides was independently conducted by 2 seasoned pathologists following the IASLC/ATS/ERS classification. The Ki-67 PI was defined as the percentage of malignant cells demonstrating positive nuclear Ki-67 immunostaining. Oncogenic driver mutations associated with LUAD were detected using real-time polymerase chain reaction assays, adhering to the NCCN Guidelines in Oncology for NSCLC. TLS quantification included TLS count (n), TLS density (n/mm²), and the TLS ratio, which is the proportion of the tumor area occupied by TLSs, ascertained through segmentation of tumor and TLS regions.^{19-21,23} To confirm the structural integrity and accuracy of TLS quantification observed in H&E stains, multiplex immunofluorescence (mIF) staining was performed, as detailed in [Appendix E1](#).²³

Statistical Analysis

Interobserver consistency for CT features was measured using intraclass correlation coefficients, 95% Bland-Altman limits, and Cohen κ statistics. Continuous variables with normal distributions were analyzed using analysis of variance and the LSD post hoc test, whereas variables not normally distributed were evaluated using the Kruskal-Wallis test. Categorical variables were analyzed using the Pearson χ^2 test or Fisher exact test, with a Bonferroni adjustment for pairwise post hoc comparisons. The diagnostic accuracy of size and density measures was assessed with receiver operating

characteristic (ROC) curves and area under the curve (AUC) statistics, and optimal cutoff points were calculated. To balance differences in clinical characteristics (age, sex, smoking history) across 3 subgroups, propensity score matching (PSM) was conducted using the 'TriMatch' R package, followed by repeated comparative analyses. The relationships between clinicoradiologic traits and the Ki-67 PI were examined using Spearman rank correlation, Mann-Whitney *U* test, and Kruskal-Wallis test. ROC and logistic regression analyses were used to identify independent predictors of EGFR mutation status (EGFR+ vs wild-type). The associations between CT characteristics and TLS parameters were explored using the Spearman correlation. For the development of a nomogram to differentiate IA from pre-IA lesions (AAH, AIS, MIA), pGGNs from Renji Hospital were randomly divided into a training set (TS, *n* = 642) and an internal validation set (IVS, *n* = 428) using a 3:2 ratio. The nomogram was built using least absolute shrinkage and selection operator—selected variables in a multivariable logistic regression model. The model's performance was evaluated in the TS, IVS, and external validation set (EVT; *n* = 159) using ROC analysis, calibration plots, and decision curve analysis (DCA). All statistical analyses were performed using R software (version 4.3.0; <https://www.r-project.org>).

RESULTS

Patients and CT Characteristics

The study included a total of 1070 pGGNs from 995 patients, with a breakdown of 18 AAH, 206 AIS, 600 MIA, and 246 IA lesions. The interobserver agreement for both quantitative and qualitative CT features was substantial to almost perfect, as detailed in [Table E1](#) and [Figure E1](#). Intraclass correlation coefficient for size and density metrics ranged from 0.84 to 0.97, whereas kappa values for categorical variables ranged from 0.80 to 0.97, indicating excellent consistency between observers.

Pathologically, the pGGNs were organized into 3 groups: the PGL group (*n* = 224, including AAH and AIS; median age 52 years; interquartile range [IQR], 40-61 years; 177 female; 10 patients who smoked), the MIA group (*n* = 600; median age 54 years; IQR, 43-64 years; 450 female; 29 patients who smoked), and the IA group (*n* = 246; median age 61 years; IQR, 52-67 years; 170 females; 9 patients who smoked). Clinical and CT feature comparisons among the PGL, MIA, and IA groups, as presented in pGGNs, are summarized in [Table E2](#). Clinically, patients identified as IA were significantly older than those in the MIA group (61 vs 54 years, *P* < .001) ([Figure E2](#)). Radiologically, significant differences were observed between PGL and MIA, as well as between MIA and IA, in terms of lesion size, density, shape, border definition, air bronchogram presence, vessel type, and voxel heterogeneity (all *P* < .001).

Prediction of Invasiveness on the Basis of Clinic-Radiologic Characteristics

[Figure 1](#), A and B, and [Table E3](#) display ROC curves and the corresponding AUC values for each size and density parameter used to differentiate among PGL, MIA, and IA groups. In the comparison of MIA versus PGL, the density parameters showed better predictive capabilities than size, with the greatest CT value achieving an AUC of 0.820. In

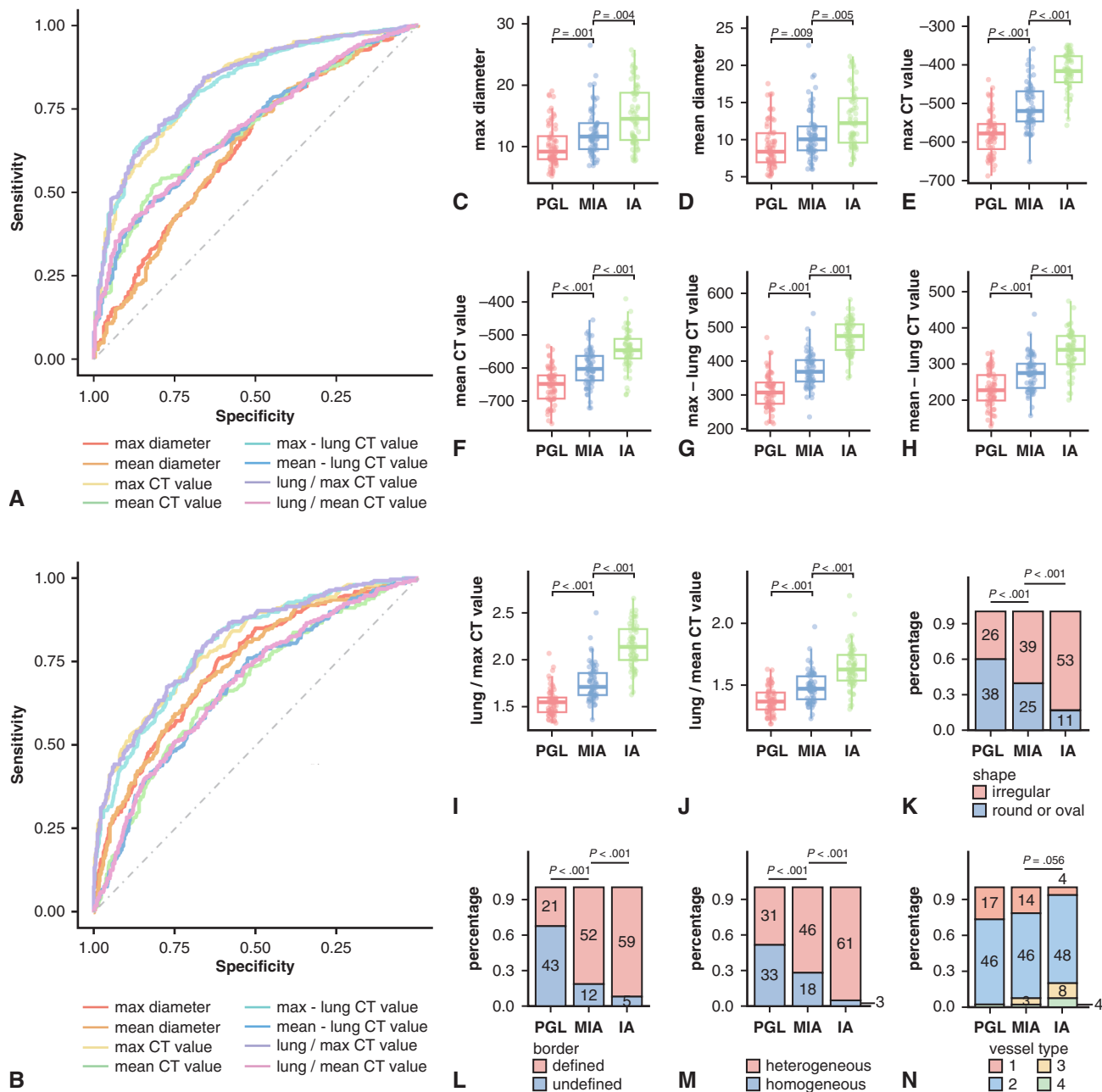


FIGURE 1. The receiver operating characteristic curve of variable parameters for differentiating PGL from MIA (A) and differentiating MIA from IA manifesting as pGGNs (B). C-N, Distribution and multigroup comparisons (PGL vs MIA, MIA vs IA) of CT characteristics after propensity score matching. C and D, The significant difference of size among pGGNs with variable invasiveness. E-J, The significant difference of parameters associated with density among PGL, MIA, and IA. K-N, Distribution and multi-group comparisons of the features of shape (K), border (L), voxel heterogeneity (M), and vessel type (N) among different invasive lesions appearing as pGGNs. The lower and upper borders of each box represent the 25th and 75th percentiles, respectively. The horizontal line inside the box indicates the median. The whiskers represent the minimum and maximum nonoutlier values, and dots outside the whiskers represent outliers. PGL, Precursor glandular lesion; MIA, minimally invasive adenocarcinoma; IA, invasive adenocarcinoma, pGGN, pure ground-glass nodule; CT, computed tomography.

addition, the relative density, calculated as the ratio of the lung to the lesion's greatest CT value (lung/max CT value), showed slightly greater accuracy, with an AUC of 0.826. The optimal cutoff values for distinguishing MIA from PGL were -548 HU for the greatest CT value and 1.62 for the lung/max CT value. The corresponding optimal

thresholds for lesion size were 8.6 mm for maximum diameter and 8.4 mm for mean diameter. In differentiating IA from MIA, both the maximum CT value and the lung/max CT value continued to exhibit strong discriminatory ability, with cutoff values of -458 HU and 1.89 , respectively. Max diameter also demonstrated significant predictive value

(AUC, 0.738), with an optimal cutoff at 11.5 mm. Overall, the lung/max CT value proved to be a reliable indicator of invasiveness, and including maximum diameter further enhanced its predictive effectiveness.

As shown in Table E4, after PSM among the 3 groups, no significant differences in clinical characteristics were observed (all $P > .1$; standardized mean difference $< .2$). All size and density parameters showed a progressive increase across the pathologic spectrum from PGL to MIA to IA, with these differences being statistically significant among the groups (all $P < .01$), as illustrated in Figure 1, C-J. Furthermore, features such as irregular shape, well-defined borders, and heterogeneous density were observed more frequently in lesions with greater degrees of invasiveness, both in comparisons between PGL and MIA, and MIA and IA (all $P < .001$) as shown in Figure 1, K-M. These results affirm that features like lesion size, density, shape, border definition, and voxel heterogeneity are crucial imaging markers for classifying invasiveness in pGGNs.

Correlation Between Clinic-Radiologic Features and Key Pathologic Characteristics

Correlation analysis showed a weak yet statistically significant positive relationship between age ($R = 0.16$, $P < .001$), max diameter ($R = 0.16$, $P < .001$), and max CT value ($R = 0.16$, $P < .001$) with the Ki-67 PI (Figure 2, A-C). Further comparative analysis revealed significant associations between greater Ki-67 PI and several radiologic features: type II/III vessels ($P < .001$), irregular shape ($P < .001$), well-defined borders ($P = .007$), air bronchogram ($P = .029$), and heterogeneous density ($P < .001$) (Figure 2, D-H, and Table E5).

A total of 371 pGGNs that underwent testing for lung cancer-associated driver gene alterations were included, with a median patient age of 58 years (IQR, 48-65 years), and the majority being female (71%, 262/371) and never-smokers (96%, 357/371). Driver mutations were identified in 77% (286 of 371) of the cases, with the most common being EGFR mutations, present in 66% (245 of 371) of cases (Figure 2, J). When comparing clinic-radiologic characteristics between EGFR-mutant (EGFR+) and wild-type (WT) pGGNs (Figure 2, K-O, Table E6), we found that EGFR+ lesions were typically found in older patients (60 vs 52 years, $P < .001$), had a larger max diameter (13.7 vs 11.0 mm, $P < .001$), greater max CT value (-454 vs -490 HU, $P < .001$), more frequently exhibited an irregular shape (79% vs 56%, $P < .001$), and displayed heterogeneous density (87% vs 72%, $P = .002$). From a histologic perspective, EGFR+ pGGNs were primarily classified as IA (62%, 151 of 245), whereas WT pGGNs were more often MIA (65%, 75 of 126) ($P < .001$). The prevalence of EGFR+ cases increased with the level of invasiveness (MIA vs AIS, $P = .005$; IA vs MIA, $P < .001$) as indicated in Figure 2,

P. Among EGFR-mutant subtypes, patients with exon 19 deletions were younger compared with those with EGFR exon 21 L858R point mutations (59 vs 63 years, $P = .002$) (Table E7). ROC and multivariate analyses pinpointed age over 54 years (odds ratio [OR], 2.082; $P = .002$), max diameter greater than 12.57 mm (OR, 2.070; $P = .011$), and max CT attenuation greater than -497.8 HU (OR, 2.378; $P = .002$) as independent predictors of EGFR mutation status in pGGNs (Tables E8 and E9). Irregular shape was also suggested as a potential predictor, although with borderline significance (OR, 1.777; $P = .052$).

Figure 3, A-C, showed a strong alignment between the radiological features of pGGNs, histologic images from H&E staining, and mIF staining. The mIF staining effectively illustrated the detailed architecture of TLSs, aligning with previous studies,²⁰ and the TLS counts were comparable with those obtained from H&E-stained slides. TLSs were observed within tumor areas, exhibiting clearly defined B-cell zones and CD4⁺ T-cell zones, with CD8⁺ T cells more diffusely distributed. Correlation analysis indicated a weak yet statistically significant relationship between TLS count and the max diameter of pGGNs ($R = 0.28$, $P = .004$), although no correlations were observed for TLS density or TLS ratio (Figure 3, D and E). Conversely, max CT value showed weak but significant positive correlations with several TLS-related metrics, including TLS count ($R = 0.31$, $P = .001$), TLS density ($R = 0.23$, $P = .019$), and TLS ratio ($R = 0.25$, $P = .009$) (Figure 3, F-H). Among other radiologic characteristics, only irregular shape was significantly linked to an increased TLS count ($P = .035$) (Table E10).

Nomogram Model for Predicting IA in pGGNs

A nomogram was created to distinguish IA from pre-IA conditions (AAH, AIS, and MIA) in patients with pGGNs, as depicted in Figure 4 (LASSO feature selection shown in Figure E3). This model calculated a total score on the basis of 4 clinic-radiologic variables: patient age, max diameter, lung/max CT value, and lesion shape. In the TS, the nomogram exhibited superior discriminatory capacity, achieving an AUC of 0.921. Calibration analysis showed a strong concordance between the predicted probabilities and actual occurrences of IA (Brier score = 0.093). DCA demonstrated that the model provided a greater net clinical benefit compared to a “treat-all” strategy within a probability threshold range of 5% to 95%. Validation in IVS maintained solid performance metrics (AUC, 0.883; Brier score = 0.102), with DCA highlighting its clinical value for probabilities between 6% and 83%. EVS from four other centers recorded an AUC of 0.890 and a Brier score of 0.132, with DCA indicating a clinical benefit for thresholds ranging from 11% to 82%. Overall, the nomogram for identifying IA displayed excellent discrimination, reliable calibration, and significant clinical net benefit.

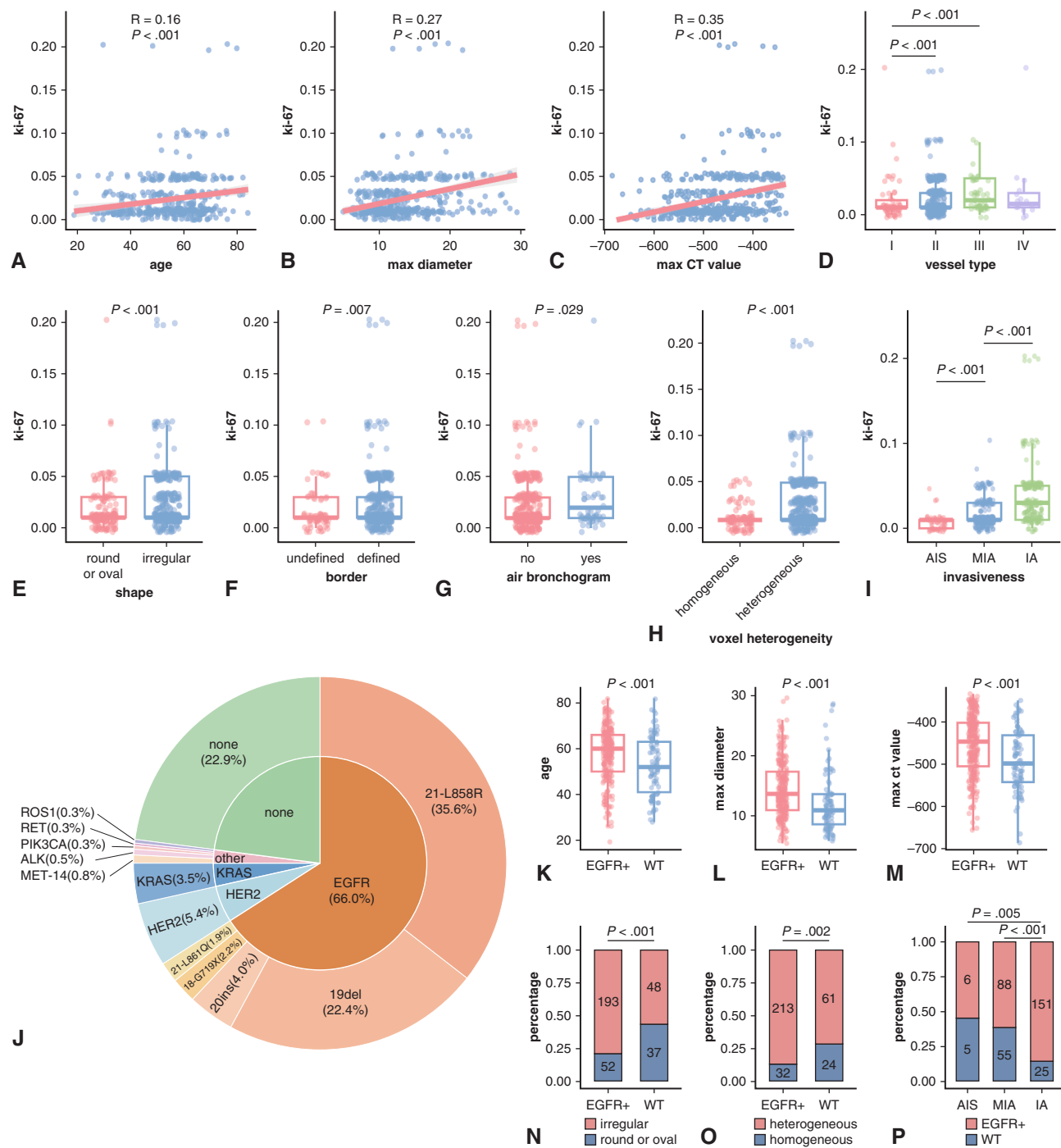


FIGURE 2. Correlation and difference analysis of Ki-67 PI and clinic-radiologic characteristics of pGGNs (A-I). A-C, The correlation between Ki-67 PI and the clinic-radiologic characteristics of age (A), max diameter (B), and max CT value (C). D-H, The differential distribution of Ki-67 PI in pGGNs with different CT features including vessel type (D), shape (E), border (F), air bronchogram (G), and voxel heterogeneity (H). I, The significant difference of Ki-67 PI in pGGNs with variable pathologic invasiveness. Distributions of driver oncogene mutations in 371 surgically resected pGGNs (J), and comparison of CT features between EGFR+ and WT pGGNs. K-O, The significant difference of age (K), max diameter (L), max CT value (M), shape (N), and voxel heterogeneity (O) between EGFR+ and WT pGGNs. P, The different proportion of EGFR+ in variable invasive lesions manifesting as pGGNs. The lower and upper borders of each box represent the 25th and 75th percentiles, respectively. The horizontal line inside the box indicates the median. The whiskers represent the minimum and maximum non-outlier values, and dots outside the whiskers represent outliers. Ki-67 PI, Ki-67 proliferation index; CT, computed tomography; pGGN, pure ground-glass nodule; EGFR, epidermal growth factor receptor; WT, wild-type.

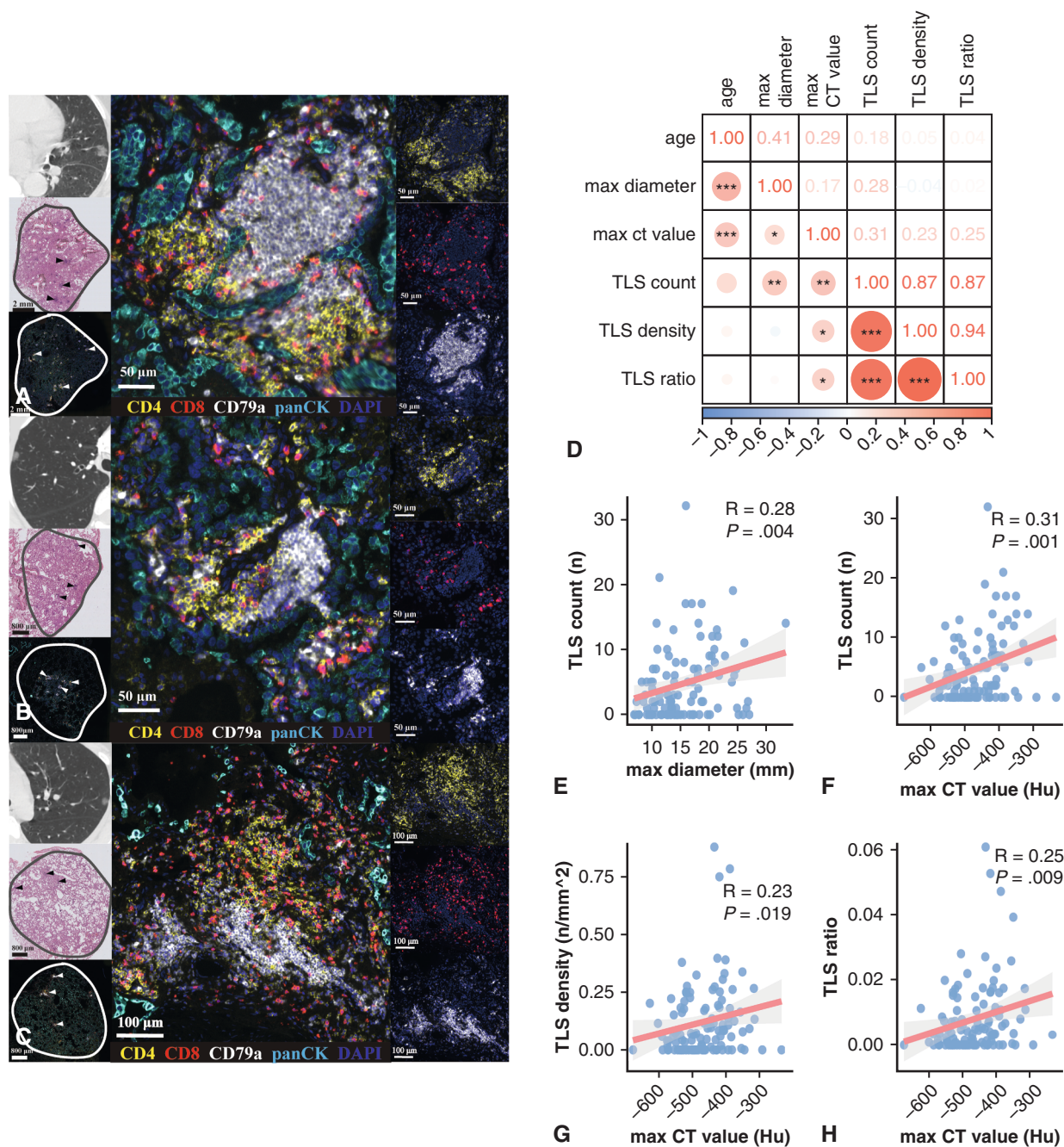


FIGURE 3. A-C, Comparison of CT images and histologic images with H&E staining and mIF staining. A, pGGNs from a 72-year-old male patient adjacent to the pleura in the upper left lobe, with 16.0 mm in max diameter and -409 HU in max CT value (HRCT image in the upper left). On the basis of the histologic image of H&E staining (middle-left image), this pGGNs was pathologically diagnosed as IA with predominantly lepidic and acinar patten. A total of 17 TLS were identified on the H&E image (partly shown by the black arrow) and confirmed on the mIF (partly shown by the white arrow on lower left image). The structure of the TLS is clearly shown on the mIF-stained image (middle image), with a central B-cell zone (stained by CD79a), a surrounding CD4+ T-cell zone, and scattered distribution of CD8+ T cells (displayed at right). B, A pGGNs from a 43-year-old female patient contact with the pleura in the upper right lobe, with 10.9 mm in max diameter and -420 HU in max CT value, which was recognized pathologically as MIA. Seven TLSs were confirmed on H&E image and mIF image. C, AIS manifesting as pGGNs from a 29-year-old male patient, with 10.2 mm in max diameter and -534 HU in max CT value. Four TLS were counted from H&E image and mIF image. D-H, Qualitative and quantitative analysis of TLSs in H&E staining and mIF staining. D, Correlation heatmap demonstrating the degree of correlation between quantitative radiographic features of pGGNs and levels of TLSs. E, Correlation analysis of TLS count and max diameter. F-H, Correlation analysis of max CT value and TLS count (F), TLS density (G), and TLS ratio (H), respectively. CT, Computed tomography; H&E, hematoxylin and eosin; mIF, multiplex immunofluorescence. pGGN, pure ground-glass nodule; HRCT, high-resolution computed tomography; IA, invasive adenocarcinoma; TLS, tertiary lymphoid structure; AIS, adenocarcinoma in situ.

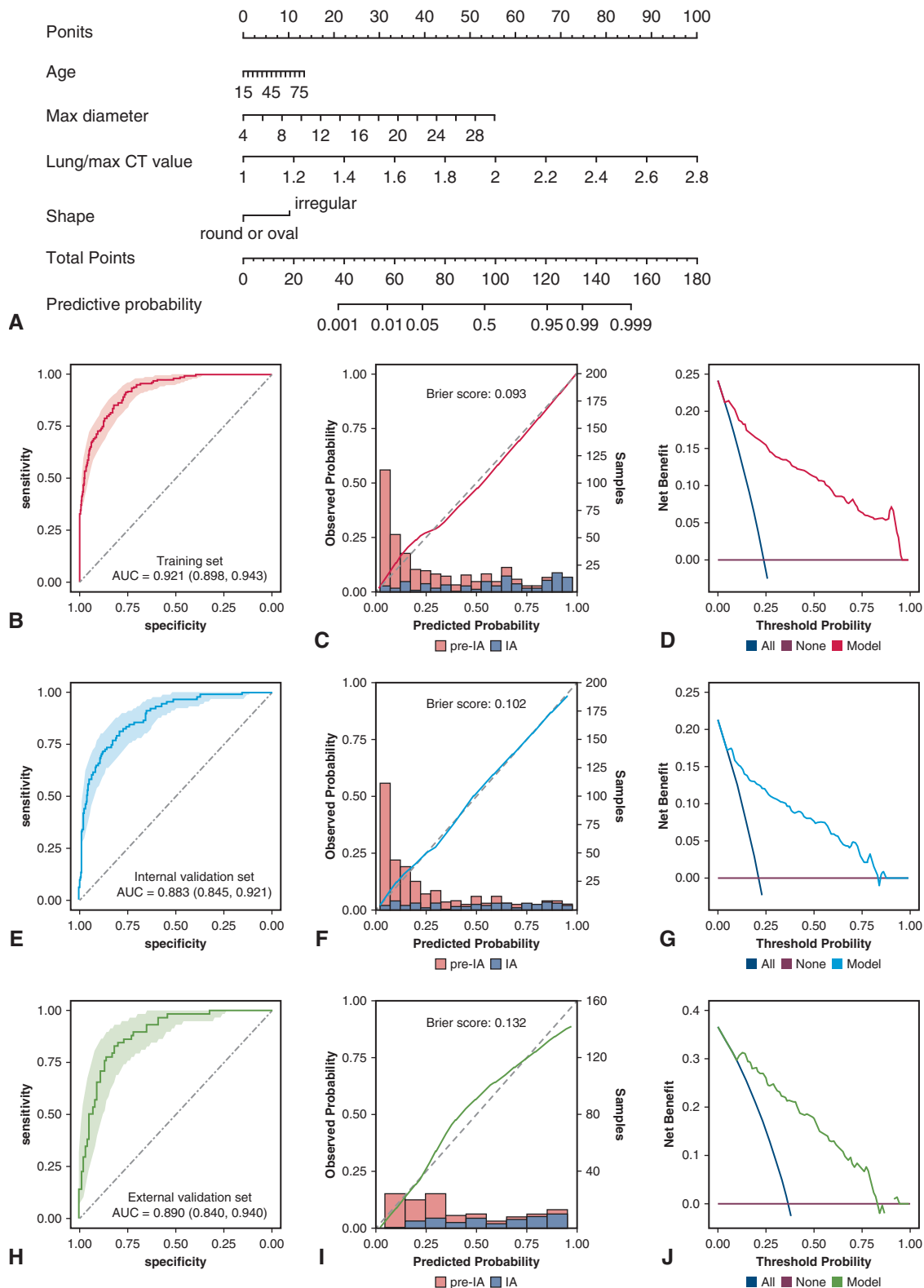


FIGURE 4. Clinical predictive modeling for discrimination of IA. A, Nomogram of the clinic-radiologic model for predicting the risk of IA. B-D, The ROC curve, calibration plot, and DCA of predictive nomogram model for differentiating IA from pre-IA in training set. E-G, The ROC curve, calibration plot, and DCA of predictive nomogram model for differentiating IA from pre-IA in internal validation set. H-J, The ROC curve, calibration plot, and DCA of predictive nomogram model for differentiating IA from pre-IA in external validation set. *IA*, Invasive adenocarcinoma; *ROC*, receiver operating characteristic; *DCA*, decision analysis.

DISCUSSION

In this study, we investigated the associations between the clinic-radiological characteristics of pGGNs and key pathologic features, including tumor invasiveness, proliferative index, driver oncogene alterations, and TLS. Consistent with previous studies,^{12,13,24} our results indicated that individuals with IA were significantly older than those with MIA. Age has consistently been recognized as an independent risk factor for lung cancer across various stages.²⁵ We suggest that this age-related trend might be linked to a decline in antitumor immune surveillance as individuals age. Consequently, lesions at similar pathologic stages like AAH or AIS may have a higher likelihood of progressing in older adults, thereby being classified as high-risk nodules that necessitate surgical intervention. Moreover, because of the slow-developing nature of pGGNs, some lesions might only manifest clinically significant characteristics requiring intervention at a more advanced age.

Size and density have been consistently identified as key independent factors for distinguishing between PGL and MIA/IA, as well as for differentiating pre-IA from IA in pGGNs.^{4,7,9,12,24} In our study, the max CT density proved to be a more effective predictor than lesion size, which has been a contentious issue in previous studies.^{7-9,26} The measurement of mean CT value may be susceptible to confounding from adjacent vessels and bronchi, resulting in potential bias. Unlike previous studies that used LDCT or conventional CT,¹⁴ our study HRCT or ultra-high-resolution computed tomography (U-HRCT) targeted scans with MPR, enhancing our ability to accurately separate tumor elements from the surrounding lung tissue and minimize measurement errors. In addition, incorporating lung parenchymal attenuation into our analysis, we observed that lung/max CT value provided superior discrimination. Previous studies have also indicated that relative and differential density metrics may be advantageous in predicting IA,^{12,27,E52} particularly in patients with emphysema.²⁸ The solid component visible on CT is generally indicative of reduced alveolar airspace, increased tissue cellularity, and thickened alveolar septa, contributing to the higher densities noted in invasive lesions. However, because of the incomplete replacement of alveolar airspaces by solid tumor components and the limited spatial resolution (200–300 μm) and partial volume effects inherent in HRCT, MIA or IA lesions may still appear radiologically as pGGNs.^{E53}

Various metrics have been utilized to evaluate the size and density of pGGNs, although the effectiveness of these parameters and their threshold values for indicating invasiveness show considerable variation across different studies (Tables E11 and E12). In our study, the nonpresence of a solid component within the nodules was verified using the lung window setting, a method corroborated by prior studies which have shown that lung window images yield a more precise correlation between CT scans and pathological assessments of solid components.²⁹ Accordingly, we

established specific cut-off values ideal for distinguishing MIA from PGL at -548 HU for max CT value, 1.62 for lung/max CT value, and 8.6 mm for max diameter. In addition, to differentiate IA from MIA, the best thresholds identified were -458 HU, 1.89 , and 11.5 mm. These thresholds are consistent with established radiologic practices.

After excluding the clinical features through PSM, we found that, beyond size and density, the characteristics of shape, border, and voxel heterogeneity showed significant differences among PGL, MIA, and IA. In line with previous research, irregular shapes and associated morphologic features such as spiculation, lobulation, the arc concave sign, and a BiA/SmA ratio ≥ 1.068 have been established as independent indicators for distinguishing MIA or IA.^{9,13,30,E54} Such irregularities in morphology can be attributed to varying proliferative dynamics within the tumor, constraints imposed by the surrounding interstitium, fibrotic changes within the lesion, or displacement of the interlobular septa.^{9,30} A well-defined border is another independent marker of MIA and IA, typically indicative of invasive growth, while a blurred margin usually suggests inflammation or AAH.^{11,12} Heterogeneous density has been reported to occur more frequently in invasive pGGNs (MIA/IA).^{10,13} Specific markers of heterogeneity including the reticulation sign, entropy, and skewness, have been identified as precise predictors of IA or radiologic progression.^{4,E55,E56} As tumors evolve, the more aggressive components that exhibit greater CT attenuation may coexist with a lepidic pattern, thereby increasing the heterogeneity observed in the scans.

Some studies have suggested that proximity to the pleura or pleural indentation may indicate invasiveness.^{12,13} Although visceral pleural invasion (VPI) is recognized as an adverse prognostic factor and categorizes lung cancer as stage IB, which may require postoperative adjuvant therapy,⁶ none of the pGGNs in our study demonstrated pathological VPI, even those in contact with the pleura. This finding aligns with prior studies.³⁰ In addition, in cases of stage I NSCLC manifested as mGGNs, pathological VPI showed no significant prognostic value.^{25,E57} Although bubble-like lucency and air bronchogram have been identified in some studies as independent predictors of IA,⁴ our study, along with the findings from Yang and colleagues²⁴ observed no significant differences in these features across varying levels of invasiveness. Moreover, Gao and colleagues^{E58} have categorized the interactions between GGNs and pulmonary vessels into 4 distinct types, with subsequent studies suggesting that vascular changes such as distortion, dilation, and type III/IV vessel alterations might serve as independent markers of IA.^{4,11-14} The presence of increased fibrosis, tumor angiogenesis, and enhanced neoplastic oxygen demand may contribute to the vascular changes noted in early-stage lung cancer.¹⁴ However, evaluating these vascular alterations is largely subjective and

relies on the expertise of the observer, often necessitating comparisons with previous imaging. In our study, vascular types III and IV did not show significant value in distinguishing between PGL, MIA, and IA. As has been observed before, vessel traversal (type II) is commonly seen in pGGNs, and both the incidence and number of intranodular vessels tend to be more closely associated with the size and anatomical position of the GGNs rather than their pathologic invasiveness.³¹

A high Ki-67 PI is closely linked with increased tumor invasiveness, the presence of high-risk histologic subtypes, poor differentiation, and more advanced stages of pathology.^{15,32} Previous studies have demonstrated a positive correlation between Ki-67 PI and patient age.¹⁵ In addition, the size of mGGNs has been proven to be an independent determinant of the Ki-67 PI.³³ Beyond size, our findings also reveal a modest but statistically significant positive correlation between max CT value and Ki-67 PI in pGGNs. This correlation might provide insights into the progression of such nodules over extended periods of observation.

In models of EGFR-mutant mice, a progression from AAH to BAC to IA was noted, highlighting the role of continuous EGFR signaling in the advancement of LUAD.^{E51} Clinically, EGFR mutations are more frequent in MIA or IA than AIS, and has been linked with the progression of pGGNs as opposed to WT.^{16,17,32,E59} Thus, pGGNs characterized by EGFR mutations were inclined to progress by AAH/AIS-MIA-IA, emphasizing the importance of more rigorous monitoring or early therapeutic interventions. Although some research has explored the relationship between size or proportion of GGNs and EGFR mutation status, results have been mixed.^{17,E60} In addition, several imaging characteristics such as bubble-like lucency, pleural retraction, and air bronchogram have been suggested as indicators of EGFR mutations in mGGNs, but their reliability in pGGNs remains uncertain.³⁴

In our study group, which predominantly consisted of female, nonsmoking Chinese patients, EGFR mutations were identified in 66% (245/371) of the pGGNs, a rate that aligns with the 64% observed in Japanese GGNs with a consolidation-to-tumor ratio greater than 50%.¹⁶ Independent predictors of EGFR mutation in pGGNs included being older than 54 years, maximum diameter >12.57 mm, and max CT value >−497.8 HU. The growing field of radiogenomics, which connects radiological “image phenotypes” with genetic “molecular phenotypes,” offers a promising new approach to the noninvasive prediction of oncogenic driver mutations, thus facilitating more tailored and effective management of lung cancer.

TLSs feature organized B-cell zones centrally and T-cell zones peripherally, mimicking the lymphoid follicles found in secondary lymphoid organs, playing a crucial role in anti-tumor immune responses.²⁰ TLS have been demonstrated to predict favorable survival outcomes in patients with early-

stage lung cancer undergoing surgical resection.^{19,20} In addition, the presence of tumor-infiltrating B cells and TLS has been linked to positive responses to immune checkpoint blockade therapy in NSCLC.¹⁸ Recent studies also suggest potential effectiveness of immune checkpoint blockade therapy in patients with multifocal GGNs.³⁵ Furthermore, strategies such as low-dose radiotherapy combined with immunotherapy, aimed at inducing TLS formation, may enhance antitumor efficacy.³⁶ Therefore, TLS can act as a significant biomarker that reflects the immune landscape of the tumor microenvironment and impacts patient prognosis.

A previous study indicated that mGGNs have a greater density of TLS compared with pGGNs.²² In our study, after normalizing by tumor area, both TLS density and TLS ratio were found to be significantly associated with max CT value, but not with lesion size. This suggests that the increased CT density in pGGNs might not only be due to more invasive tumor components but could also reflect a higher infiltration of immune cells associated with tumor invasion, potentially leading to the development of additional TLSs. Although our study did not directly assess the relationship between TLS and pathologic invasiveness, recent findings suggest that TLS count, density, and ratio may escalate with the invasiveness.²⁰

pGGNs are linked to an outstanding postoperative prognosis, with reported 5-year rates for recurrence-free survival, disease-free survival, and overall survival approaching 100%.²⁵ Pathologically, the 5-year recurrence-free survival and disease-free survival rates for MIA are 100%, whereas for IA manifesting as pGGNs, they remain high at 97.7%.^{E61} In a long-term follow-up study of pGGNs (diameter range, 3–20 mm), approximately 10% of nodules showed progression after a median follow-up of 59 months, with a median volume doubling time of 769 days.^{E62} For mGGNs, long-term studies have indicated that a conservative management approach is viable until there is radiological evidence of growth in the solid component, without adversely affecting lung cancer-specific mortality.³⁷ Importantly, lung function can be preserved during this surveillance period, ensuring that patients remain candidates for surgical intervention should more aggressive lesions develop in other areas of the lung.³⁷ These observations collectively advocate for a strategy of delayed surgical intervention for pGGNs until radiologic signs strongly suggest IA.

Wang and colleagues³⁸ and She and colleagues³⁹ previously developed preoperative nomogram models to predict the invasiveness (MIA and IA) of pGGNs, achieving excellent discriminatory capabilities with reported AUCs between 0.916 and 0.970. In our current study, we have refined this approach by constructing a nomogram specifically designed to distinguish IA from PGL and MIA. This model includes 4 easily obtainable clinic-radiologic factors—patient age, max diameter, lung/max CT value,

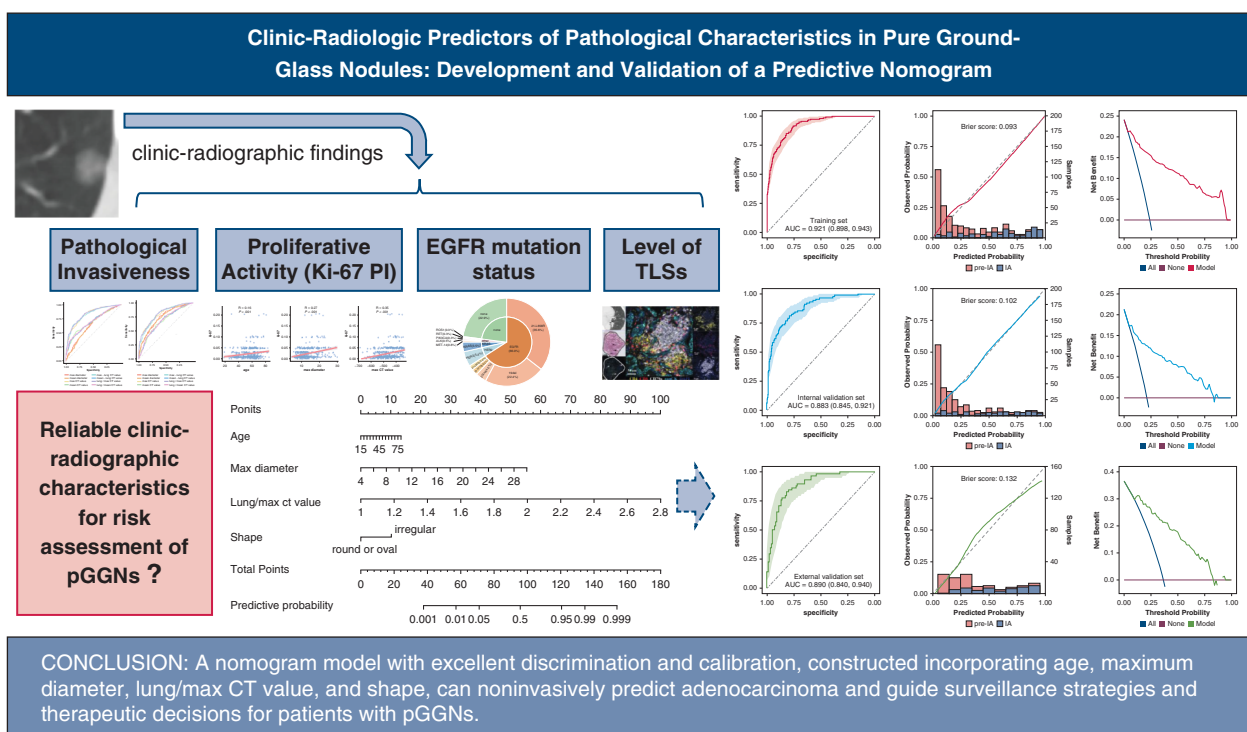


FIGURE 5. This study highlights the diagnostic value of CT-based size and density metrics in predicting invasiveness, proliferative activity, EGFR mutations, and TLS levels in pGGNs. A predictive nomogram integrating clinical and radiologic features (age, maximum diameter, lung/max CT value, and shape) demonstrated excellent discrimination, calibration, and clinical net benefit across the training, internal validation, and external validation cohorts. CT, Computed tomography; EGFR, epidermal growth factor receptor; TLS, tertiary lymphoid structure; pGGN, pure ground-glass nodule; Ki-67 PI, Ki-67 proliferation index.

and shape. Our nomogram exhibited outstanding discrimination, calibration, and net clinical benefit across various decision thresholds in both IVS and EVS. Notably, the EVS encompassed 159 pGGNs from 140 patients at 4 medical centers, including Yunnan Province, which reports one of the highest rates of incidence and mortality from NSCLC in China.⁴⁰ This multi-institutional validation underscores the model's broad applicability and adaptability in different geographic and epidemiologic contexts.

Several limitations of this study should be acknowledged. First, because of its retrospective cross-sectional design, there is an inherent risk of selection bias because only surgically removed pGGNs were analyzed, excluding those that were not resected or were non-neoplastic. Second, the study relied on a single preoperative CT scan, which does not allow for a longitudinal analysis of nodule development. Third, inconsistencies in CT-acquisition protocols between different institutions could have led to variations in the quantitative measurements obtained. Finally, the dependence on visual assessment and manual measurements may compromise the reproducibility of the results, unlike what might be achieved with automated, computer-aided quantification methods.

CONCLUSIONS

This study elucidated the relationships between the radiologic characteristics of pGGNs, specifically size and density,

and crucial pathologic features such as invasiveness, proliferative activity, oncogenic mutations, and TLS. By developing a comprehensive nomogram that integrates patient age, max diameter, lung/max CT value, and shape, we identified essential predictors of IA. This model has proven to be an effective noninvasive tool, enhancing the accuracy of diagnosis and facilitating tailored treatment strategies for patients with pGGNs (as summarized in Figure 5).

Conflict of Interest Statement

The authors reported no conflicts of interest.

The *Journal* policy requires editors and reviewers to disclose conflicts of interest and to decline handling or reviewing manuscripts for which they may have a conflict of interest. The editors and reviewers of this article have no conflicts of interest.

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Key Words: pure ground-glass nodule (pGGN), lung adenocarcinoma (LUAD), tertiary lymphoid structures (TLSs), EGFR mutation, Ki-67 proliferation index, nomogram

APPENDIX E1: METHODS

Patients

Inclusion criteria for pure ground-glass nodules (pGGNs) were as follows: (1) pGGN lesions were surgically resected via lobectomy or sublobar resection and pathologically confirmed as atypical adenomatous hyperplasia (AAH), adenocarcinoma in situ (AIS), minimally invasive adenocarcinoma (MIA), or invasive adenocarcinoma (IA); (2) the interval between computed tomography (CT) imaging and surgery was less than 1 month; and (3) lesions were identified as single or multiple (≤ 3), and the max diameter of the pGGNs was ≤ 3 cm in greatest dimension. Patients with the following conditions were excluded from the study: (1) patients lacking complete and reliable clinical-pathologic information; (2) benign lesions confirmed pathologically, or nonstage IA lung adenocarcinoma according to the ninth edition TNM staging^{E1}; (3) no high-resolution CT (HRCT) or U-HRCT target scanning images with a thickness of ≤ 1 mm in a month. Personal information, medical history, and operative records of patients with pGGNs were obtained from an electronic medical record system.

CT Technique

CT of chest was performed using equipment from 2 manufacturers: Philips Healthcare and GE HealthCare. CT scanning parameters were as follows: tube voltage, 120 kVp; tube current, 80 to 298 mA; scanning slice thickness, 5 mm; reconstruction slice thickness and interval, 1 mm; matrix: 512×512 ; rotation time, 0.5 second. Scanning of patients with their upper limbs raised in the supine position was performed at the end of deep inspiration and completed within 1 breath hold. The thoracic CT covers the area from the thoracic inlet to the costophrenic angle. The CT features were analyzed in the lung window setting (window level, -600 HU; width, 1600 HU) in picture archiving and communication system. Distinct from low-dose computed tomography used in lung cancer screening, the confirmation of pGGNs and the acquisition of CT features are based on nonenhanced, thin-layer HRCT or U-HRCT target scanning to accurately determine the absence of a solid component and to avoid, as far as possible, the visual interference of vessels or bronchioles.^{E2} The most recent preoperative CT scan meeting the aforementioned criteria (generally performed within one month before surgery) was reviewed for each patient.

Analysis of CT Features

First, multiplanar reconstruction analyses (axial, sagittal, and coronal) were performed in the lung window setting to avoid the interference of vessels, bronchioles, or fibrous streak. The pGGNs without any solid component (opacity that completely obscured the lung parenchyma) confirmed by 2 observers (a radiologist with 15 years of experience

in chest CT diagnosis and a thoracic surgeon with 10 years of clinical experience) in parallel were included in the study.^{E3} Then the following CT features were assessed and measured by each of 2 observers: (1) lesion size: maximal diameter and vertical diameter (in millimeter), (2) lesion density: maximal and mean CT attenuation value (in Hounsfield unit, HU), (3) CT attenuation value of lung parenchyma (HU), (4) lesion location: right upper lobe, right middle lobe, right lower lobe, left upper lobe, and left lower lobe, (5) lesion multiplicity (solitary, multiple), (f) presence of pleural attachment, (7) shape (irregular, round, or oval), (8) border (defined, undefined), (9) presence of air bronchogram, (10) presence of bubble-like lucency, (11) vessel type in the interior or periphery of the lesion (I, II, III, IV), and (12) voxel heterogeneity (heterogeneous, homogenous). The max diameter of pGGNs was measured in axial greatest dimension and the max perpendicular diameter was measured in the same layer. The mean diameter = (the max diameter + the perpendicular diameter)/2.^{E4} Given that the physical density of lung nodules (g/cm^3) is linearly related to the CT attenuation value (HU), we used various CT values to characterize the density.^{E5} A region of interest, ovoid or irregular, covering one half to two thirds of the largest area in a pGGNs away from areas of bubble-like lucency, vessel, bronchus, or stripes of fiber was selected.^{E6} And automatic measurements are used to obtain the max CT value and mean CT value. The CT attenuation value of lung parenchyma (lung CT value) is the average of the measurements of three regions of interest at different axial layers, which are captured in the normal lung parenchyma region without lung-marking within the same lung lobes as the pGGNs. The difference density (max-lung CT value, mean-lung CT value) was the max or mean CT value minus the lung CT value.^{E7} The relative density was calculated by dividing the lung CT value by the max or mean CT value.^{E7-E9} Pleural attachment was lesion attached or connected by streaks with pleural surface including pleural tag or pleural indentation described previously.^{E8,E10-E12} Air bronchogram sign was an air-filled (low-attenuation) bronchi was found inside the lesions (high-attenuation).^{E13,E14} Differently, bubble-like lucency, or vacuole sign, manifest as small spots of air attenuation within lesions.^{E3,E6} The spatial relationships between GGNs and supplying blood vessels were categorized into 4 types: I, vessels passing by GGNs; II, intact vessels passing through GGNs; III, distorted, dilated or tortuous vessels seen within GGNs; IV, more complicated vasculature other than described previously.^{E15}

Analysis of Pathology

Histologic subtype was classified according to IASLC/ATS/ERS classification of lung adenocarcinoma.^{E16} AAH is a localized, small (usually diameter < 5 mm) lesion due

to proliferation of mildly to moderately atypical type II pneumocytes (Clara cells) lining alveolar walls or respiratory bronchioles.^{E16} AIS is defined a localized small (≤ 3 cm) adenocarcinoma with growth restricted to neoplastic cells along preexisting alveolar structures (lepidic growth), lacking stromal, vascular, or pleural invasion.^{E16} MIA is a small adenocarcinoma (≤ 3 cm), with a predominantly lepidic pattern and ≤ 5 mm invasive component in greatest dimension in any one focus, excluded by invasion of lymphatics, blood vessels, or pleura, and contain of tumor necrosis,^{E16} whereas IA represents adenocarcinomas with the diameter of max invasive component >5 mm.^{E16} As an additional note, the invasive component is defined as (1) histologic subtypes other than a lepidic pattern (ie, acinar, papillary, micropapillary, solid and/or mucinous component) or (2) myofibroblastic stroma associated with invasive tumor cells.^{E16,E17} The percentage of each histologic pattern was recorded in 5% to 10% increments to determine the predominant histologic pattern (subtype) and quantify any patterns to determine the tumor grade.^{E17} Also, lymph nodes sampled intraoperatively were assessed pathologically to exclude patients with non-N0-staged lung cancer. Data were collected from clinicopathological reports, and the original pathological slides were not re-examined in this study.

Analysis of Genetic Alterations

Lung adenocarcinoma–related oncogenic driver mutations were determined using real-time polymerase chain reaction if formalin-fixed paraffin-embedded (FFPE) tissue was available and informed consent was obtained, and amplification refractory mutation system technique was used to detect the mutation status of epidermal growth factor receptor (EGFR) exons 18, 19, 20, and 21.^{E12} The panel of the test included EGFR, KRAS, ALK rearrangement, ROS1 rearrangement, BRAF, NTRK gene fusion, RET rearrangement, MET ex14 skipping mutations (MET-14), ERBB2 (HER2), and PIK3CA, corresponding with the recommendations of the National Comprehensive Cancer Network Guidelines for non–small cell lung cancer.^{E18} The gene mutation detection kit was purchased from Amoy Diagnostics Co, Ltd (Xiamen).

Analysis of Tertiary Lymphoid Structures (TLS)

TLS were identified on H&E-stained FFPE sections; the definition of a TLS required presence of a germinal center-like structure with high density of lymphocytes with poorly visible cytoplasm either within the tumoral area or in direct contact with the tumoral cells on the margin of the tumors.^{E19-E21} Then, segmentation of TLS and tumors was performed for the identification of the features: the number of TLS, the area of TLS, and tumors regions.^{E22,E23} TLS count (n) was identified either within the tumoral area or in direct contact with the tumoral cells

on the margin of the tumors.^{E19} Subsequently, TLS density was calculated as the number of TLS (n/mm^2) divided by the area of tumor, and the TLS ratio represented the sum of the area of the TLS divided by the area of tumor.^{E19,E23-E26}

TLS were qualified and quantified using H&E, and multiplex immunofluorescence (mIF) staining is used to confirm the structure of TLS and verify the accuracy of the number of TLS counted through H&E.^{E19} mIF was performed on 4- μm sections of FFPE tumor samples from our clinical studies, after deparaffinization, rehydration, epitope retrieval, and blocking.^{E21} Sections were incubated in targeted antibodies—including anti-CD4 (1:100, #48274, CST), anti-CD8 (1:400, #85336, CST), anti-CD79a (1:200, #48750, CST), and anti-pan-cytokeratin (Kit-0009, MXB Biotechnologies)—detected with Opal Polymer horseradish peroxidase anti-rabbit/mouse, and amplified with fluorescent reagents. Slides were counterstained with 4,6-diamidino-2-phenylindole. The mIF kit was purchased from Akoya Biosciences (America). Images of fluorescent slides were acquired on a PhenoImager HT.

Statistical Analysis

Interobserver agreement for CT findings of pGGNs was investigated by using the intraclass correlation coefficient and 95% Bland-Altman limits of agreement for continuous variables and Cohen Kappa statistic for qualitative features. Intraclass correlation coefficient >0.80 is considered to be good agreement between the 2 observers. Kappa values of 0.61–0.80 are substantial and 0.81–1.00 represents almost-perfect agreement.^{E27-E29} Patient clinic-demographics and CT findings were compared among 3 histopathologic diagnoses (PGL, MIA, and IA). Normal continuous variables were expressed as mean $\pm$ standard deviation and analyzed by analysis of variance and least significant difference *t* test. Otherwise, non-normal continuous variables, described as median (interquartile range), were evaluated using Kruskal-Wallis test. Categorical variables expressed as numbers (percentages) were compared using Pearson χ^2 test (when no cell of a contingency table has expected count less than 5) or Fisher exact *t* tests (when any cell of a contingency table has expected count less than 5). Post hoc pairwise comparisons were corrected using the Bonferroni method. In our study, the internal correlation of multiple pGGNs in the same patients was not considered, as multiple GGNs are considered to be primary lesions, which need to be evaluated separately when treated.^{E30-E33} The area under curve (AUC) of the receiver operating characteristic (ROC) curve was used to evaluate the predictive performance of the main quantitative radiologic features (including variable parameters of size and density). The binomial exact method was used to determine the confidence intervals of AUCs. The optimal cut-off threshold was delimited

according to the Youden index of ROC analysis, and the corresponding sensitivity and specificity were calculated. Previous studies have suggested differences in clinical characteristics among various invasive lesions, so to clarify the role of CT findings in identifying PGL, MIA, and IA, we performed the propensity score match of 3 subgroups using the 'TriMatch' R package to balance differences of clinical characteristics (including age, sex, and smoking history) among subgroups. After propensity score match, analyses of difference and post hoc comparison among different invasive subgroups were again performed. To investigate the correlation between clinical-imaging characteristics and Ki-67 proliferation index (Ki-67 PI), Spearman rank correlation analysis, Mann-Whitney *U* test, and Kruskal-Wallis test were used. The clinical and CT features were compared between the EGFR mutation (EGFR+) group and wild-type group, between the EGFR exon 21 L858R point mutations subtype and exon 19 deletions subtype. Quantitative variables were analyzed with the Mann-Whitney *U* test or unpaired *t* test. The Pearson χ^2 test or Fisher exact test was used to compare qualitative variables. Cut-off values for quantitative features with significant difference ($P < .05$) were determined by ROC analysis. To identify independent variables that could be used in differentiating EGFR+ from wild-type pGGNs, logistic regression analysis was conducted. Characteristics with a *P* value of less than .05 at differential analysis were used as the input variables for multiple logistic regression analysis. A backward stepwise selection mode was used, with iterative entry of variables on the basis of test results ($P < .05$). The removal of variables was based on likelihood ratio statistics with a probability of 0.10.^{E34} Spearman rank correlation analysis was used to explore the correlation between clinic-radiologic features and TLS level.

To build a nomogram model to differentiate IA from pre-IA lesions (AAH, AIS, and MIA), the pGGNs from Renji hospital were divided randomly into the training set (TS, *n* = 642) and internal validation set (IVS, *n* = 428) at a ratio of 3:2. Note that different nodules from the same individual are not required to be assigned to the same set. The least absolute shrinkage and selection operator was used to select the most useful predictive features in the TS. Univariable logistic regression analysis was performed to evaluate clinical and radiologic variables. The significant clinical and radiologic risk factors were then imported into a stepwise multivariable logistic regression analysis to develop the nomogram. Subsequently, the performance of the nomogram was assessed with discrimination, calibration, and clinical benefit and evaluated on the validation sets. The IVS and EVT (*n* = 159) from multiple centers were used to test the nomogram model. The ROC analysis and AUC were used to evaluate the discrimination of the models in

TS, IVS and EVT. The calibration plot visualized the reliability of the predicted probability comparing the probability and actual proportions.^{E35} A decision curve analysis (DCA) was performed for evaluating the potential clinical diagnostic effects of model.^{E36} Least absolute shrinkage and selection operator logistic regression was performed using the "glmnet" package. The "rms" R package was used to build a nomogram and calibration plot. The ROCs were plotted using the "pROC" package. The DCA was performed with the "ggDCA" package. All statistical analysis was performed with R software (version 4.3.0; <https://www.r-project.org>).

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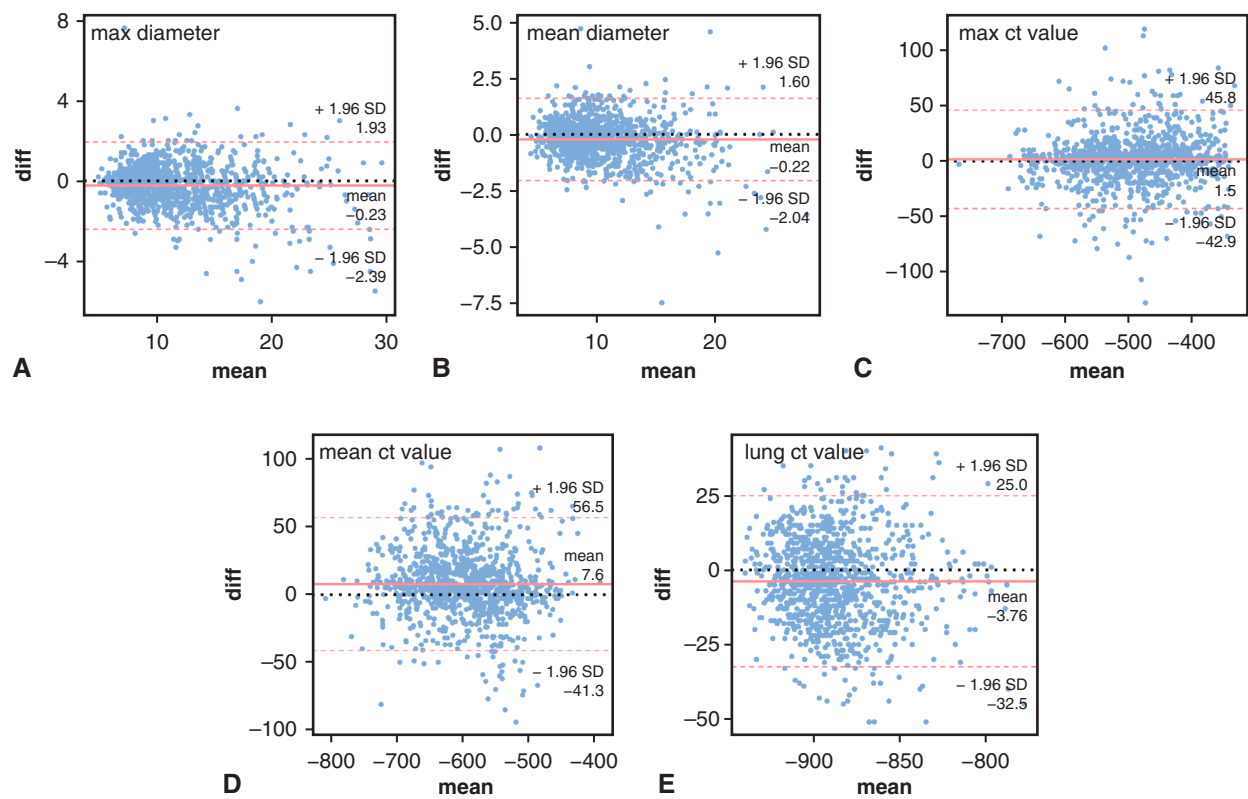


FIGURE E1. Bland-Altman plots showing interobserver agreement of measurements of the max diameter (A), mean diameter (B), max CT value (C), mean CT value (D), and lung CT value (E) of pGGNs. *X axes* show mean measurements and *Y axes* show differences between measurements of 2 observers. *CT*, Computed tomography; *pGGN*, pure ground-glass nodule.

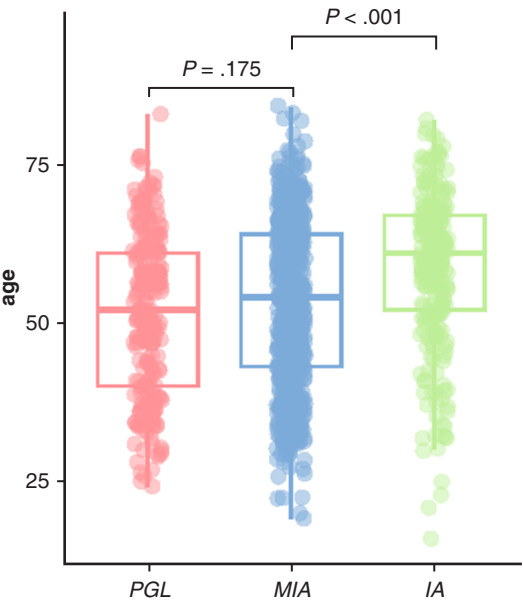


FIGURE E2. Differences in the age of patients with variable invasive lesions appearing pGGNs. *pGGN*, Pure ground-glass nodule; *PGL*, precursor glandular lesion; *MIA*, minimally invasive adenocarcinoma; *IA*, invasive adenocarcinoma.

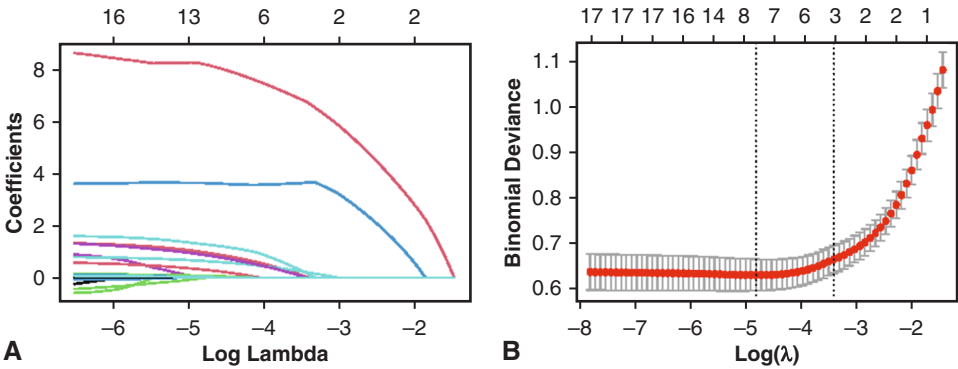


FIGURE E3. Clinic-radiologic features selection using LASSO regression model. A, LASSO coefficient profiles of 17 clinic-radiologic characteristics. B, Selection of the optimal tuning parameters by cross-validation. *LASSO*, Least absolute shrinkage and selection operator.

TABLE E1. Interobserver agreement for CT features

CT features	ICC	Kappa
Max diameter	0.967	—
Mean diameter	0.966	—
Max CT value	0.953	—
Mean CT value	0.924	—
Lung CT value	0.848	—
Pleural attachment	—	0.965
Shape	—	0.928
Border	—	0.805
Air bronchogram	—	0.878
Bubble-like lucency	—	0.907
Vessel type	—	0.916
Voxel heterogeneity	—	0.813

CT, Computed tomography; ICC, intraclass correlation coefficient; kappa, kappa coefficient for categorical variables.

TABLE E2. Comparisons of groups in terms of clinical and CT features

Variables	PGL (n = 224)	MIA (n = 600)	IA (n = 246)	P	P1	P2	SMD
Sex, n (%)				.044	.798	.282	0.152
Female	177 (79)	450 (75)	170 (69)				
Male	47 (21)	150 (25)	76 (31)				
Age, y, median (IQR)	52 (40, 61)	54 (43, 64)	61 (52, 67)	<.001	.175	<.001	0.373
Smoking history, n (%)				.755	1	1	0.039
No	214 (96)	571 (95)	237 (96)				
Yes	10 (4)	29 (5)	9 (4)				
Max diameter, median (IQR)	8.96 (7.7, 11.16)	10.4 (8.75, 13.45)	14.21 (11.56, 18.62)	<.001	<.001	<.001	0.872
Mean diameter, median (IQR)	8.01 (7.04, 9.91)	9.32 (7.88, 11.45)	12.32 (9.91, 15.74)	<.001	<.001	<.001	0.839
Max CT value, mean $\pm$ SD	-568.35 $\pm$ 54.48	-496.07 $\pm$ 58.32	-427.76 $\pm$ 56.23	<.001	<.001	<.001	1.671
Mean CT value, mean $\pm$ SD	-638.81 $\pm$ 53.6	-596.61 $\pm$ 62.86	-559.23 $\pm$ 61.1	<.001	<.001	<.001	0.903
Lung CT value, mean $\pm$ SD	-888.26 $\pm$ 25.49	-886.8 $\pm$ 26.82	-886.9 $\pm$ 26.33	.77	1.000	1.000	0.037
Max-lung CT value, mean $\pm$ SD	319.92 $\pm$ 52.81	390.73 $\pm$ 58.92	459.14 $\pm$ 56.97	<.001	<.001	<.001	1.66
Mean-lung CT value, mean $\pm$ SD	249.45 $\pm$ 51.02	290.19 $\pm$ 60.54	327.67 $\pm$ 59	<.001	<.001	<.001	0.924
Lung/max CT value, mean $\pm$ SD	1.58 $\pm$ 0.15	1.81 $\pm$ 0.21	2.11 $\pm$ 0.26	<.001	<.001	<.001	1.661
Lung/mean CT value, mean $\pm$ SD	1.4 $\pm$ 0.11	1.5 $\pm$ 0.16	1.6 $\pm$ 0.17	<.001	<.001	<.001	0.939
Pleural attachment, n (%)				.366	.993	1.000	0.086
No	154 (69)	389 (65)	154 (63)				
Yes	70 (31)	211 (35)	92 (37)				
Shape, n (%)				<.001	<.001	<.001	0.602
Irregular	95 (42)	346 (58)	204 (83)				
Round or oval	129 (58)	254 (42)	42 (17)				
Border, n (%)				<.001	<.001	<.001	0.78
Defined	105 (47)	510 (85)	230 (93)				
Undefined	119 (53)	90 (15)	16 (7)				
Air bronchogram, n (%)				<.001	<.001	<.001	0.449
No	220 (98)	553 (92)	192 (78)				
Yes	4 (2)	47 (8)	54 (22)				
Bubble-like lucency, n (%)				.01	.017	.22	0.142
No	198 (88)	566 (94)	223 (91)				
Yes	26 (12)	34 (6)	23 (9)				
Vessel type, n (%)				<.001	<.001	<.001	0.517
I	71 (32)	108 (18)	24 (10)				
II	147 (66)	449 (75)	173 (70)				
III	5 (2)	34 (6)	33 (13)				
IV	1 (0)	9 (2)	16 (7)				
Voxel heterogeneity, n (%)				<.001	<.001	<.001	0.652
Heterogeneous	119 (53)	447 (74)	228 (93)				
Homogeneous	105 (47)	153 (26)	18 (7)				
Lobe, n (%)				.064	1.000	.020	0.201
LLL	25 (11)	79 (13)	22 (9)				
LUL	66 (29)	155 (26)	85 (35)				
RLL	27 (12)	76 (13)	39 (16)				
RML	15 (7)	38 (6)	21 (9)				
RUL	91 (41)	252 (42)	79 (32)				

(Continued)

TABLE E2. Continued

Variables	PGL (n = 224)	MIA (n = 600)	IA (n = 246)	<i>P</i>	<i>P1</i>	<i>P2</i>	SMD
Pathology, n (%)				<.001	<.001	<.001	NA
AAH	18 (8)	0 (0)	0 (0)				
AIS	206 (92)	0 (0)	0 (0)				
MIA	0 (0)	600 (100)	0 (0)				
IA	0 (0)	0 (0)	246 (100)				

Significant differences are indicated in bold. *P* indicates the *P* value of comparisons among PGL, MIA and IA; *P1* indicates the *P* value of post hoc comparisons between PGL and MIA, adjusted by Bonferroni; and *P2* indicates the *P* value of post hoc comparisons between MIA and IA, adjusted by Bonferroni; *PGL*, Precursor glandular lesions (including AAH and AIS); *MIA*, minimally invasive adenocarcinoma; *IA*, invasive adenocarcinoma; *SMD*, standardized mean difference; *IQR*, interquartile range; *CT*, computed tomography; *SD*, standard deviation; *LLL*, left lower lobe, *LUL*, left upper lobe, *RLL*, right lower lobe, *RML*, right middle lobe, *RUL*, right upper lobe; *NA*, not applicable; *AAH*, atypical adenomatous hyperplasia; *AIS*, adenocarcinoma in situ.

TABLE E3. AUC of ROC curve and cut-off value of diameter and CT value for differentiating among PGL, MIA, and IA

Variables	Diagnostic performance for distinguishing MIA from PGL				Diagnostic performance for distinguishing IA from MIA			
	AUC (95%CI)	Cut-off value	Sensitivity	Specificity	AUC (95% CI)	Cut-off value	Sensitivity	Specificity
Max diameter	0.637 (0.594, 0.68)	8.6 mm	0.765	0.460	0.738 (0.701-0.774)	11.5 mm	0.760	0.613
Mean diameter	0.635 (0.591, 0.678)	8.4 mm	0.657	0.567	0.736 (0.699- 0.772)	10.6 mm	0.691	0.668
Max CT value	0.820 (0.789, 0.852)	−548 HU	0.827	0.665	0.799 (0.765-0.832)	−458 HU	0.703	0.733
Mean CT value	0.693 (0.655, 0.731)	−594HU	0.518	0.817	0.671 (0.631-0.711)	−568 HU	0.610	0.677
Max-lung CT value	0.819 (0.788, 0.850)	367 HU	0.650	0.853	0.796 (0.763-0.829)	414 HU	0.797	0.662
Mean-lung CT value	0.692 (0.653, 0.730)	295 HU	0.470	0.826	0.675 (0.636-0.715)	293 HU	0.760	0.525
Lung/max CT value	0.826 (0.796, 0.857)	1.62	0.843	0.661	0.805 (0.772- 0.837)	1.89	0.780	0.683
Lung/mean CT value	0.695 (0.658, 0.733)	1.52	0.430	0.871	0.678 (0.639- 0.718)	1.53	0.675	0.602

MIA, Minimally invasive adenocarcinoma; *PGL*, precursor glandular lesions (including AAH and AIS); *IA*, invasive adenocarcinoma; *AUC*, area under the curve; *CT*, computed tomography.

TABLE E4. Comparisons of groups in terms of clinical and CT features after propensity score matching

Variables	PGL (n = 64)	MIA (n = 64)	IA (n = 64)	P	P1	P2	SMD
Sex, n (%)				1	1.000	1.000	<0.001
Female	41 (64)	41 (64)	41 (64)				
Male	23 (36)	23 (36)	23 (36)				
Age, y, median (IQR)	55.5 (44.75, 65)	55.5 (44.75, 65)	56 (46.75, 65.25)	.919	1.000	1.000	0.042
Smoking history, n (%)				.514	1	1	0.138
No	62 (97)	62 (97)	59 (92)				
Yes	2 (3)	2 (3)	5 (8)				
Max diameter, median (IQR)	9.18 (7.93, 11.7)	11.62 (9.57, 13.81)	14.52 (11.07, 18.76)	<.001	.001	.004	0.791
Mean diameter, median (IQR)	8.4 (6.98, 10.88)	10.05 (8.57, 11.79)	12.23 (9.61, 15.58)	<.001	.009	.005	0.704
Max CT value, mean $\pm$ SD	-580.21 $\pm$ 53.52	-508.12 $\pm$ 56.04	-419.41 $\pm$ 47.46	<.001	<.001	<.001	2.068
Mean CT value, mean $\pm$ SD	-652.31 $\pm$ 52.76	-599 $\pm$ 56.08	-546.81 $\pm$ 55.53	<.001	<.001	<.001	1.287
Lung CT value, mean $\pm$ SD	-890.07 $\pm$ 25.67	-879.97 $\pm$ 27.39	-888.4 $\pm$ 22.03	.054	.072	.178	0.263
Max-lung CT value, mean $\pm$ SD	309.86 $\pm$ 51.9	371.85 $\pm$ 50.55	468.99 $\pm$ 49.26	<.001	<.001	<.001	2.1
Mean-lung CT value, mean $\pm$ SD	237.76 $\pm$ 49.1	280.97 $\pm$ 48.13	341.59 $\pm$ 54.45	<.001	<.001	<.001	1.357
Lung/max CT value, mean $\pm$ SD	1.55 $\pm$ 0.15	1.75 $\pm$ 0.19	2.14 $\pm$ 0.23	<.001	<.001	<.001	2.044
Lung/mean CT value, mean $\pm$ SD	1.37 $\pm$ 0.1	1.48 $\pm$ 0.13	1.64 $\pm$ 0.17	<.001	<.001	<.001	1.318
Pleural attachment, n (%)				.291	1.000	.528	0.184
No	46 (72)	49 (77)	41 (64)				
Yes	18 (28)	15 (23)	23 (36)				
Shape, n (%)				<.001	<.001	<.001	0.627
Irregular	26 (41)	39 (61)	53 (83)				
Round or oval	38 (59)	25 (39)	11 (17)				
Border, n (%)				<.001	<.001	<.001	1
Defined	21 (33)	52 (81)	59 (92)				
Undefined	43 (67)	12 (19)	5 (8)				
Air bronchogram, n (%)				.002	.36	.408	0.43
No	63 (98)	58 (91)	51 (80)				
Yes	1 (2)	6 (9)	13 (20)				
Bubble-like lucency, n (%)				.33	.726	1	0.175
No	55 (86)	60 (94)	58 (91)				
Yes	9 (14)	4 (6)	6 (9)				
Vessel type, n (%)				<.001	1	.056	0.568
I	17 (27)	14 (22)	4 (6)				
II	46 (72)	46 (72)	48 (75)				
III	0 (0)	3 (5)	8 (12)				
IV	1 (2)	1 (2)	4 (6)				
Voxel heterogeneity, n (%)				<.001	<.001	<.001	0.794
Heterogeneous	31 (48)	46 (72)	61 (95)				
Homogeneous	33 (52)	18 (28)	3 (5)				
Lobe, n (%)				.199	1.000	.120	0.4
LLL	7 (11)	6 (9)	7 (11)				
LUL	17 (27)	12 (19)	25 (39)				
RLL	8 (12)	10 (16)	12 (19)				
RML	3 (5)	3 (5)	1 (2)				
RUL	29 (45)	33 (52)	19 (30)				
Pathology, n (%)				<.001	<.001	<.001	NaN
AAH	14 (22)	0 (0)	0 (0)				
AIS	50 (78)	0 (0)	0 (0)				

(Continued)

TABLE E4. Continued

Variables	PGL (n = 64)	MIA (n = 64)	IA (n = 64)	P	P1	P2	SMD
MIA	0 (0)	64 (100)	0 (0)				
IA	0 (0)	0 (0)	64 (100)				

The significant differences are indicated in bold. *P* indicates the *P* value of comparisons among PGL, MIA and IA; *P*1 indicates the *P* value of post hoc comparisons between PGL and MIA, adjusted by Bonferroni; *P*2 indicates the *P* value of post hoc comparisons between MIA and IA, adjusted by Bonferroni. *PGL*, Precursor glandular lesions (including AAH and AIS); *MIA*, minimally invasive adenocarcinoma; *IA*, invasive adenocarcinoma; *SMD*, standardized mean differences; *CT*, computed tomography; *LLL*, left lower lobe, *LUL*, left upper lobe, *RLL*, right lower lobe, *RML*, right middle lobe, *RUL*, right upper lobe; *NaN*, not a number (SMD could not be calculated due to zero variance in one or more groups); *AAH*, atypical adenomatous hyperplasia; *AIS*, adenocarcinoma in situ.

TABLE E5. Comparison of Ki-67 PI in pGGNs with different clinic-radiologic features

Variables	Total n (%)	Ki-67 PI	
		Median (IQR)	P value
Sex			.114
Female	330 (69)	0.01 (0.01, 0.03)	
Male	148 (31)	0.01 (0.01, 0.05)	
Smoking history			.469
No	451 (94)	0.01 (0.01, 0.03)	
Yes	27 (6)	0.01 (0.005, 0.04)	
Pleural attachment			.389
No	310 (65)	0.01 (0.01, 0.03)	
Yes	168 (35)	0.01 (0.01, 0.05)	
Shape			<.001
Irregular	323 (68)	0.01 (0.01, 0.05)	
Round or oval	155 (32)	0.01 (0.01, 0.03)	
Border			.007
Defined	408 (85)	0.01 (0.01, 0.03)	
Undefined	70 (15)	0.01 (0.01, 0.03)	
Air bronchogram			.029
No	414 (87)	0.01 (0.01, 0.03)	
Yes	64 (13)	0.02 (0.01, 0.05)	
Bubble-like lucency			.545
No	441 (92)	0.01 (0.01, 0.03)	
Yes	37 (8)	0.01 (0.01, 0.03)	
Vessel type			.013
I	61 (13)	0.01 (0.01, 0.02)	
II	352 (74)	0.01 (0.01, 0.03)	
III	49 (10)	0.02 (0.01, 0.05)	
IV	16 (3)	0.015 (0.01, 0.03)	
Voxel heterogeneity			<.001
Heterogeneous	383 (80)	0.01 (0.01, 0.05)	
Homogeneous	95 (20)	0.01 (0.01, 0.01)	
Lobe			.517
LLL	54 (11)	0.01 (0.01, 0.03)	
LUL	152 (32)	0.01 (0.01, 0.03)	
RLL	65 (13)	0.01 (0.01, 0.03)	
RML	27 (6)	0.01 (0.01, 0.05)	
RUL	180 (38)	0.01 (0.01, 0.03)	
Pathology			<.001
AIS	50 (10)	0.01 (0.01, 0.01)	
MIA	247 (52)	0.01 (0.01, 0.03)	
IA	181 (38)	0.03 (0.01, 0.05)	

The significant differences are indicated in bold. *Ki-67 PI*, Ki-67 proliferation index; *LLL*, left lower lobe, *LUL*, left upper lobe, *RLL*, right lower lobe, *RML*, right middle lobe, *RUL*, right upper lobe; *AIS*, adenocarcinoma in situ; *MIA*, minimally invasive adenocarcinoma; *IA*, invasive adenocarcinoma.

TABLE E6. Comparisons of groups (EGFR mutation and WT) in terms of clinical and CT features

Variables	Total (n = 330)	EGFR+ (n = 245)	WT (n = 85)	P
Sex, n (%)				.879
Female	237 (72)	177 (72)	60 (71)	
Male	93 (28)	68 (28)	25 (29)	
Age, y, median (IQR)	59 (48, 65.75)	60 (50, 66)	52 (41, 63)	<.001
Smoking history, n (%)				.159
No	319 (97)	239 (98)	80 (94)	
Yes	11 (3)	6 (2)	5 (6)	
Max diameter, median (IQR)	13.1 (10.45, 16.85)	13.7 (11, 17.35)	10.95 (8.65, 13.65)	<.001
Max CT value, mean $\pm$ SD	-463.5 $\pm$ 71.87	-454.35 $\pm$ 70.06	-489.9 $\pm$ 70.87	<.001
Pleural attachment, n (%)				.627
No	224 (68)	164 (67)	60 (71)	
Yes	106 (32)	81 (33)	25 (29)	
Shape, n (%)				<.001
Irregular	241 (73)	193 (79)	48 (56)	
Round or oval	89 (27)	52 (21)	37 (44)	
Border, n (%)				.666
Defined	286 (87)	214 (87)	72 (85)	
Undefined	44 (13)	31 (13)	13 (15)	
Air bronchogram, n (%)				.07
No	281 (85)	203 (83)	78 (92)	
Yes	49 (15)	42 (17)	7 (8)	
Bubble-like lucency, n (%)				.854
No	310 (94)	231 (94)	79 (93)	
Yes	20 (6)	14 (6)	6 (7)	
Vessel type, n (%)				.219
I	36 (11)	23 (9)	13 (15)	
II	248 (75)	184 (75)	64 (75)	
III	33 (10)	26 (11)	7 (8)	
IV	13 (4)	12 (5)	1 (1)	
Voxel heterogeneity, n (%)				.002
Heterogeneous	274 (83)	213 (87)	61 (72)	
Homogeneous	56 (17)	32 (13)	24 (28)	
Lobe, n (%)				.283
LLL	32 (10)	28 (11)	4 (5)	
LUL	102 (31)	76 (31)	26 (31)	
RLL	46 (14)	36 (15)	10 (12)	
RML	20 (6)	15 (6)	5 (6)	
RUL	130 (39)	90 (37)	40 (47)	
Pathology, n (%)				<.001
AIS	11 (3)	6 (2)	5 (6)	
MIA	143 (43)	88 (36)	55 (65)	
IA	176 (53)	151 (62)	25 (29)	

The significant differences are indicated in bold. *EGFR+*, Epidermal growth factor receptor mutation status; *WT*, wild type; *IQR*, interquartile range; *CT*, computed tomography; *SD*, standard deviation; *LLL*, left lower lobe; *LUL*, left upper lobe; *RLL*, right lower lobe; *RML*, right middle lobe; *RUL*, right upper lobe; *AIS*, adenocarcinoma in situ, *MIA*, minimally invasive adenocarcinoma; *IA*, invasive adenocarcinoma.

TABLE E7. Comparisons of subtype of EGFR mutation (19del and L858R point mutation) in terms of clinical and CT features

Variables	Total (n = 215)	19del (n = 83)	L858R (n = 132)	P
Sex, n (%)				.467
Female	160 (74)	59 (71)	101 (77)	
Male	55 (26)	24 (29)	31 (23)	
Age, y, median (IQR)	61 (53, 66.5)	59 (48, 64)	62.5 (54.75, 68)	.002
Smoking history, n (%)				1
No	212 (99)	82 (99)	130 (98)	
Yes	3 (1)	1 (1)	2 (2)	
Max diameter, median (IQR)	14.05 (11.55, 17.66)	13.4 (11.16, 17.2)	14.3 (11.64, 18.2)	.373
Max CT value, mean ± SD	−453.31 ± 70.4	−452.08 ± 69.77	−454.08 ± 71.04	.839
Pleural attachment, n (%)				1
No	148 (69)	57 (69)	91 (69)	
Yes	67 (31)	26 (31)	41 (31)	
Shape, n (%)				.338
Irregular	173 (80)	70 (84)	103 (78)	
Round or oval	42 (20)	13 (16)	29 (22)	
Border, n (%)				.558
Defined	184 (86)	73 (88)	111 (84)	
Undefined	31 (14)	10 (12)	21 (16)	
Air bronchogram, n (%)				1
No	176 (82)	68 (82)	108 (82)	
Yes	39 (18)	15 (18)	24 (18)	
Air cavities, n (%)				1
No	203 (94)	78 (94)	125 (95)	
Yes	12 (6)	5 (6)	7 (5)	
Vessel type, n (%)				1
I	12 (6)	4 (5)	8 (6)	
II	168 (78)	65 (78)	103 (78)	
III	25 (12)	10 (12)	15 (11)	
IV	10 (5)	4 (5)	6 (5)	
Voxel heterogeneity, n (%)				1
Heterogeneous	189 (88)	73 (88)	116 (88)	
Homogeneous	26 (12)	10 (12)	16 (12)	
Lobe, n (%)				.172
LLL	22 (10)	10 (12)	12 (9)	
LUL	67 (31)	30 (36)	37 (28)	
RLL	33 (15)	15 (18)	18 (14)	
RML	13 (6)	2 (2)	11 (8)	
RUL	80 (37)	26 (31)	54 (41)	
Pathology, n (%)				.632
AIS	3 (1)	2 (2)	1 (1)	
MIA	77 (36)	28 (34)	49 (37)	
IA	135 (63)	53 (64)	82 (62)	

The significant differences are indicated in bold. *19del*, EGFR exon 19 deletion; *L858R*, EGFR exon 21 L858R point mutation; *CT*, computed tomography; *LLL*, left lower lobe; *LUL*, left upper lobe; *RLL*, right lower lobe; *RML*, right middle lobe; *RUL*, right upper lobe; *AIS*, adenocarcinoma in situ, *MIA*, minimally invasive adenocarcinoma; *IA*, invasive adenocarcinoma.

TABLE E8. The optimal cut-off value of age, max diameter, and max CT value for predicting EGFR mutation

Variables	Cut-off value	Sensitivity	Specificity
Age, y	54.5	66.90%	57.60%
Max diameter, mm	12.57	62.40%	65.90%
Max CT value, HU	−497.75	73.90%	50.60%

CT, Computed tomography; EGFR, epidermal growth factor receptor.

TABLE E9. Multivariable binary logistic regression analysis of clinical and CT features in predicting EGFR mutation of pure ground-glass nodule

Variable	Odds ratio (95% CI)	P value
(Intercept)	0.532 (0.297-0.942)	.032
Age (>54 y)	2.082 (1.212-3.589)	.008
Max diameter (>12.57 mm)	2.07 (1.178-3.657)	.012
Max CT value (>−497.8 HU)	2.378 (1.384-4.091)	.002
Shape—irregular	1.777 (0.989-3.168)	.052

CT, Computed tomography; EGFR, epidermal growth factor receptor; CI, confidence interval.

TABLE E10. Comparison of TLSs levels in pGGNs with different clinic-radiologic features

Variables	Total n (%)	TLS count		TLS density		TLS ratio	
		Median (IQR)	P	Median (IQR)	P	Median (IQR)	P
Sex			.274		.763		.97
Female	81 (76)	2 (0, 7)		0.071 (0, 0.179)		0.004 (0, 0.011)	
Male	25 (24)	6 (0, 10)		0.066 (0, 0.249)		0.003 (0, 0.015)	
Smoking history			.305		.343		.693
No	102 (96)	2.5 (0, 7)		0.068 (0, 0.181)		0.004 (0, 0.011)	
Yes	4 (4)	6 (5, 6.5)		0.102 (0.089, 0.144)		0.004 (0.003, 0.008)	
Pleural attachment			.992		.782		.802
No	68 (64)	2 (0, 7.25)		0.062 (0, 0.177)		0.004 (0, 0.011)	
Yes	38 (36)	3.5 (0, 7)		0.078 (0, 0.208)		0.004 (0, 0.011)	
Shape			.035		.345		.117
Irregular	62 (58)	5 (1, 9)		0.08 (0.014, 0.183)		0.005 (0.001, 0.012)	
Round Or Oval	44 (42)	2 (0, 5)		0.055 (0, 0.173)		0.002 (0, 0.009)	
Border			.18		.304		.329
Defined	95 (90)	3 (0, 7.5)		0.074 (0, 0.183)		0.004 (0, 0.012)	
Undefined	11 (10)	1 (0, 4)		0.046 (0, 0.147)		0.001 (0, 0.009)	
Air bronchogram			.265		.699		.816
No	82 (77)	2.5 (0, 7)		0.072 (0, 0.19)		0.004 (0, 0.014)	
Yes	24 (23)	4 (0.75, 10.25)		0.059 (0.009, 0.177)		0.004 (0.001, 0.009)	
Bubble-like lucency			.852		.949		.887
No	99 (93)	3 (0, 7)		0.071 (0, 0.181)		0.004 (0, 0.011)	
Yes	7 (7)	2 (0.5, 9)		0.071 (0.01, 0.181)		0.004 (0.001, 0.01)	
Vessel type			.295		.3		.292
I	2 (2)	4.5 (4.25, 4.75)		0.253 (0.217, 0.288)		0.015 (0.011, 0.018)	
II	81 (76)	2 (0, 7)		0.065 (0, 0.179)		0.003 (0, 0.011)	
III	17 (16)	3 (0, 9)		0.085 (0, 0.125)		0.005 (0, 0.010)	
IV	6 (6)	9.5 (4, 15.75)		0.124 (0.052, 0.237)		0.012 (0.004, 0.019)	
Voxel heterogeneity			.531		.834		.717
Heterogeneous	90 (85)	2.5 (0, 8)		0.072 (0, 0.183)		0.004 (0, 0.011)	
Homogeneous	16 (15)	3 (1.5, 5)		0.062 (0.025, 0.173)		0.003 (0.001, 0.008)	
Pathology			.01		.483		.487
AIS	7 (7)	0 (0, 2.5)		0 (0, 0.126)		0 (0, 0.007)	
MIA	35 (33)	2 (0, 4)		0.046 (0, 0.216)		0.003 (0, 0.012)	
IA	64 (60)	5 (0, 10)		0.082 (0, 0.172)		0.005 (0, 0.012)	

The significant differences are indicated in bold. *TLS*, Tertiary lymphoid structure; *AIS*, adenocarcinoma in situ; *MIA*, minimally invasive adenocarcinoma; *IA*, invasive adenocarcinoma.

TABLE E11. ROC analysis of diameter used in previous studies for distinguishing pGGNs with different invasiveness

Study	Year	Groups and number of pGGNs	CT feature	Cut-off value	Sensitivity	Specificity	AUC
Eguchi et al ^{E37}	2014	AIS:MIA/IA = 47:54	Max diameter	11 mm	98.50%	46.80%	0.75
Jin et al ^{E38}	2015	AAH/AIS:MIA/IA = 21:73	Max diameter	10.5 mm	86.30%	61.90%	0.71
Qi et al ^{E7}	2019	benign/AAH/AIS:MIA/IA = 123:132	Max diameter	6.5 mm	69.70%	30.10%	0.78
Fu et al ^{E39}	2021	AAH/AIS:MIA/IA = 141:165	Max diameter	10.1 mm	68.50%	62.40%	0.67
Liang et al ^{E40}	2023	benign/AAH/AIS:MIA/IA = 486:390	Mean diameter	10 mm	63.10%	81.50%	NA
Han et al ^{E41}	2017	AAH/AIS:MIA: IA = 102:61	Max diameter	17.2 mm	82.00%	67.70%	0.78
Qi et al ^{E7}	2019	benign/AAH/AIS:MIA:IA = 231:24	Max diameter	11.5 mm	75.00%	91.80%	0.86
Jiang et al ^{E9}	2021	MIA:IA = 47:53	Max diameter	13.9 mm	79.20%	55.30%	0.72
Chu et al ^{E6}	2019	AAH/AIS:MIA: IA = 141:31	Max diameter	10.5 mm	87.10%	70.90%	0.84
Fu et al ^{E42}	2020	AIS:MIA:IA = 331:101	Max diameter	10.5 mm	88.10%	71.90%	0.89
Yang et al ^{E43}	2020	AIS:MIA:IA = 523:136	Max diameter	10.8 mm	81.00%	79.00%	0.85
Yang et al ^{E43}	2020	AIS:MIA:IA = 523:136	Mean diameter	10.1 mm	86.00%	75.00%	0.85
Liang et al ^{E44}	2023	Benign/AAH/AIS:MIA:IA = 736:140	Mean diameter	10.5 mm	87.90%	74.70%	NA
Chang et al ^{E45}	2023	Benign/AAH/AIS:MIA:IA = 167:75	Mean diameter	6.9 mm	76.00%	57.50%	0.71

pGGN, Pure ground-glass nodule; CT, computed tomography; AUC, the area under the curve of receiver operating characteristic curve; AIS, adenocarcinoma in situ; MIA, minimally invasive adenocarcinoma; IA, invasive adenocarcinoma; AAH, atypical adenomatous hyperplasia.

TABLE E12. ROC analysis of CT value used in previous studies for distinguishing pGGNs with different invasiveness

First author	Year	Groups and number of pGGNs	CT feature	Cut-off value	Sensitivity	Specificity	AUC
Eguchi et al ^{E37}	2014	AIS:MIA/IA = 47:54	Mean CT value	−680 HU	95.80%	35.10%	0.77
She et al ^{E46}	2017	AAH/AIS: MIA/IA = 501:397	Mean CT value	−560 HU	NA	NA	NA
Chu et al ^{E6}	2019	AAH/AIS: MIA/IA = 73:99	Mean CT value	−632 HU	78.80%	59.80%	0.72
Liang et al ^{E40}	2023	benign/AAH/AIS:MIA/IA = 486:390	Mean CT value	−631 HU	51.00%	80.90%	NA
Qi et al ^{E7}	2019	benign/AAH/AIS:MIA/IA = 123:132	max CT value	−532 HU	57.60%	86.20%	0.72
Ichinose et al ^{E47}	2020	AAH/AIS: MIA/IA = 117:63	max CT value	−300 HU	83.00%	88.00%	0.86
Fu et al ^{E39}	2021	AAH/AIS: MIA/IA = 141:165	max CT value	−502 HU	68.50%	70.20%	0.73
Qi et al ^{E7}	2019	benign/AAH/AIS:MIA/IA = 123:132	relative CT value (lung/max)	1.57	66.70%	82.90%	0.78
Han et al ^{E41}	2017	AAH/AIS/MIA: IA = 102:61	Mean CT value	−617 HU	70.50%	53.90%	0.63
Yang et al ^{E43}	2020	AIS/MIA: IA = 523:136	Mean CT value	−582 HU	66.00%	60.00%	0.66
Jiang et al ^{E9}	2021	MIA:IA = 47:53	Mean CT value	−597 HU	69.80%	80.00%	0.67
Liang et al ^{E44}	2023	Benign/AAH/AIS/MIA:IA = 736:140	Mean CT value	−621 HU	69.30%	77.50%	NA
Chang et al ^{E45}	2023	Benign/AAH/AIS/MIA:IA = 167:75	Mean CT value	−567 HU	78.70%	47.90%	0.64
Qi et al ^{E7}	2019	Benign/AAH/AIS/MIA:IA = 231:24	Max CT value	−575 HU	95.80%	57.10%	0.78
Zhao et al ^{E8}	2018	AIS/MIA:IA = 68:47	Relative CT value (lung/mean)	1.6	72.30%	64.70%	NA
Qi et al ^{E7}	2019	Benign/AAH/AIS/MIA:IA = 231:24	Relative CT value (lung/max)	1.54	91.70%	60.20%	0.78
Jiang et al ^{E9}	2021	MIA:IA = 47:53	Relative CT value (lung/mean)	1.43	71.70%	55.30%	0.62

pGGN, Pure ground-glass nodule; CT, computed tomography; AUC, the area under the curve of receiver operating characteristic curves; AIS, adenocarcinoma in situ; MIA, minimally invasive adenocarcinoma; IA, invasive adenocarcinoma; AAH, atypical adenomatous hyperplasia; NA, not applicable.